

**Dubai Metro, UAE
Jebel Ali Depot**

**Structural Design Calculation
(Enhanced Design)
Building 9 – Traction Power Station**

July 2007

Client: JT Metro JV	Contract No. (if any):
Project Title: Dubai Metro, UAE	Project No.: JRD4561
Document No.: DM001-E-AAM-CDP-DR-DCC-373909	Controlled Copy No.:
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Covering Letter / Transmittal Ref. No.: To be advised	Date of Issue: 21 Jul 2007

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A1	Internal Review	CK Lam Oct 06	P. Shepherd-Smith Oct 06	B. Maney Oct 06
Revision	Description	Prepared by / date	Reviewed by / date	App. or Auth. by / date

Distribution (if insufficient space, please use separate paper)

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1	JT Metro JV
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Note: App. and Auth. mean "Approved" and "Authorized" respectively.

Table of Contents

1	INTRODUCTION	001
2	ROOF SIDE FRAME ANALYSIS AND DESIGN	003
3	ROOF BEAMS AND SIDE BEAM DESIGN.....	037
4	CONNECTIONS DESIGN	052

1 INTRODUCTION

Document No.: **DM001-E-AAM-CDP-DR-DCC-373909**

Rev A4

Date: 21 Jul 2007

Page

INTRODUCTION

This calculation report is submitted as a supplementary calculation to cover all new additions, amendments and updates to previously submitted report.

Please refer to Document no. DM001-E-AAM-CDP-DR-DCC-373909 Rev A3 for previously submitted calculations and details

Document No.: **DM001-E-AAM-CDP-DR-DCC-373909**

Rev A4

Date: **21 Jul 2007**

Page

2 ROOF SIDE FRAME ANALYSIS AND DESIGN

Document No.: **DM001-E-AAM-CDP-DR-DCC-373909**

Rev A4

Date: 21 Jul 2007

Page



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Job Title

Client

Job No

Sheet No

Rev

1

Part

Ref

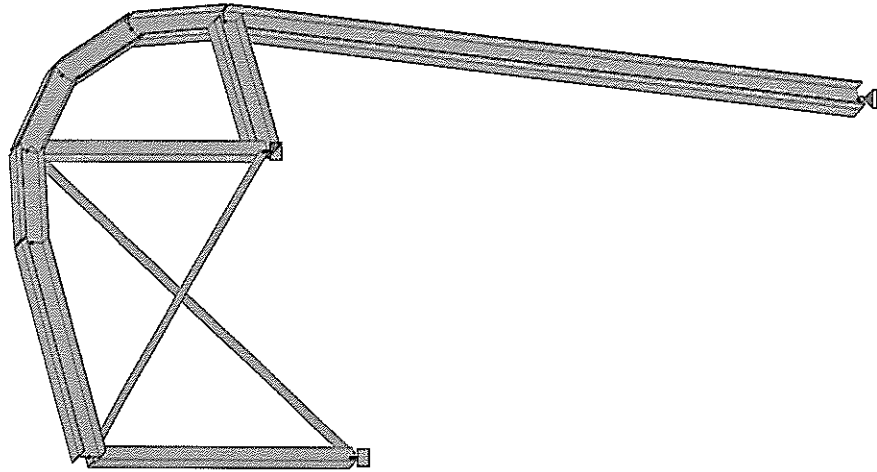
By

Date JULY 2007

Chd

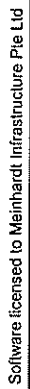
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Date/Time 09-Jul-2007 16:29



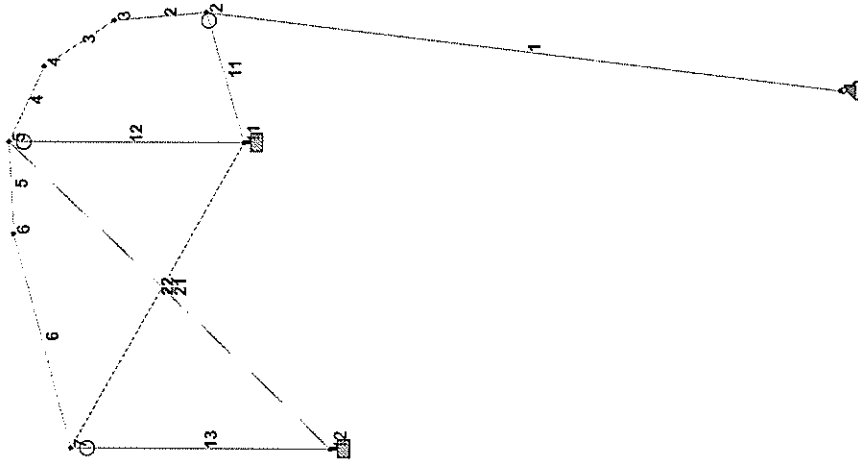
TPS BUILDING - SIDE FRAME

Load 1

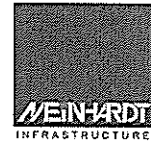


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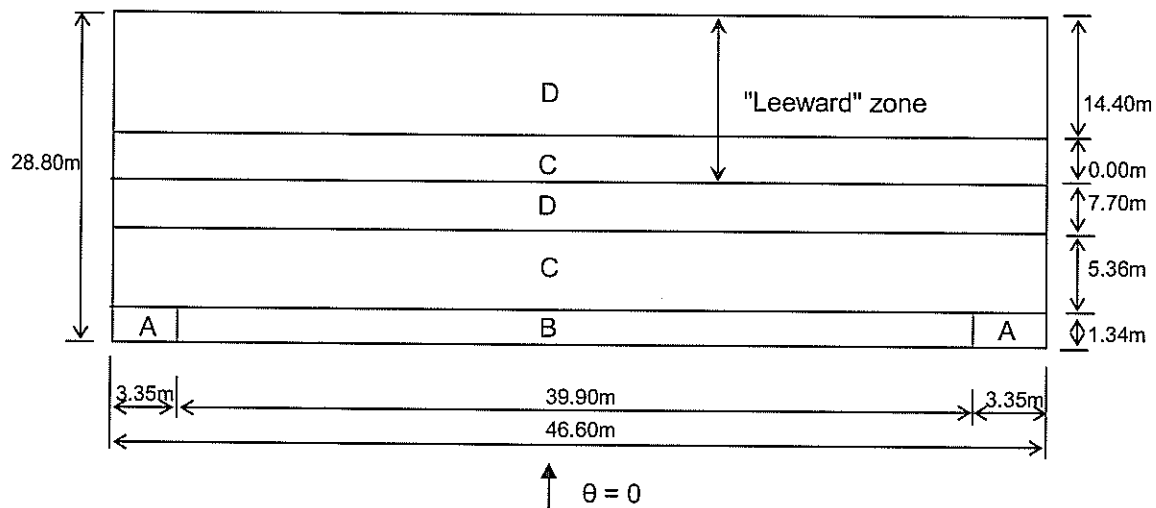
1000



50, N7 2 MEMBER ACIDENDES



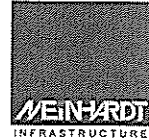
Site:	Dubai Metro Project	Date:	9-Jul-07	Job No.:	68010.01
	TPS Building				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to eave)
PLAN


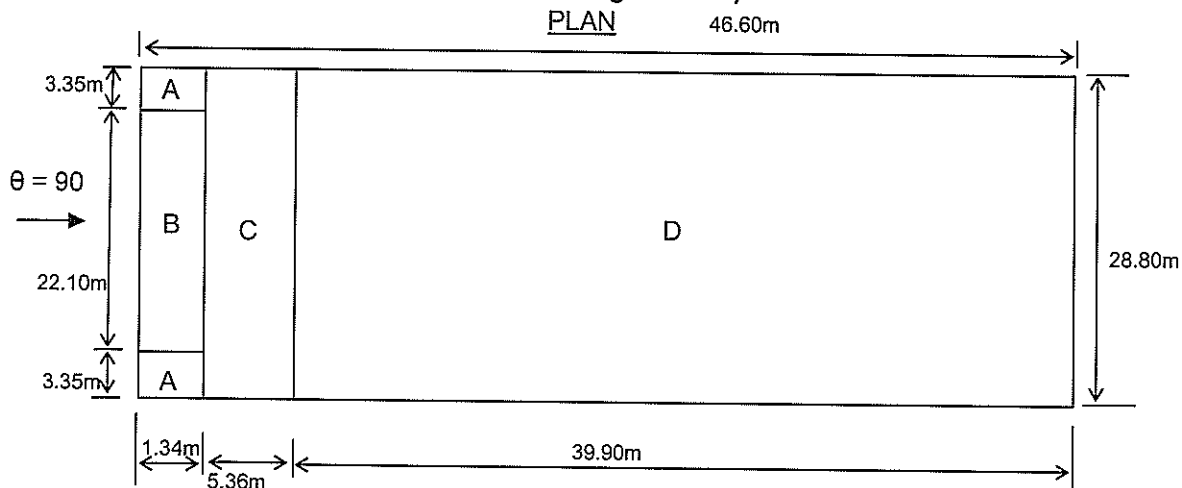
Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 46.6 \text{ m} \\
 H &= 6.7 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 2 \times 6.7 \\
 &= 13.4 \text{ m}
 \end{aligned}$$

Zone	Windward		Leeward		
	Area (m ²)	Cpe	Area	Cpe1	Cpe2
A	9.0	-2			
B	53.5	-1.4			
C	249.8	-0.7	0.0	-0.7	-0.7
D	358.8	-0.2	671.0	-0.2	0.2
Total	671.0		671.0		
Average		-0.51		-0.20	0.20



Site:	Dubai Metro Project	Date:	9-Jul-07	Job No.:	68010.01
	TPS Building				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to gable end)


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 28.8 \text{ m} \\
 H &= 6.7 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 2 \times 6.7 \\
 &= 13.4 \text{ m}
 \end{aligned}$$

For maximum uplift, assume zone C coefficient =>

$$C_{pe} = -0.7$$

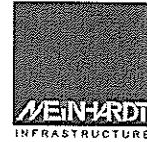
For maximum downward force, assume zone D coefficient =>

$$C_{pe} = 0.2$$

Wind Load Computation

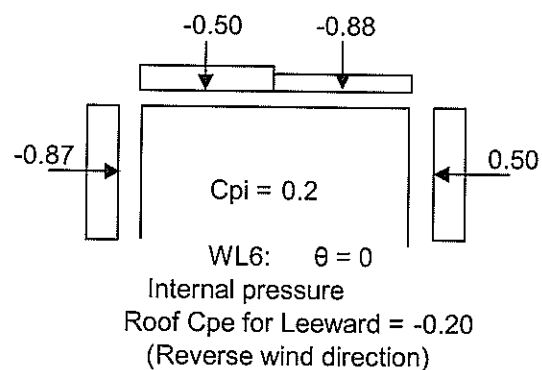
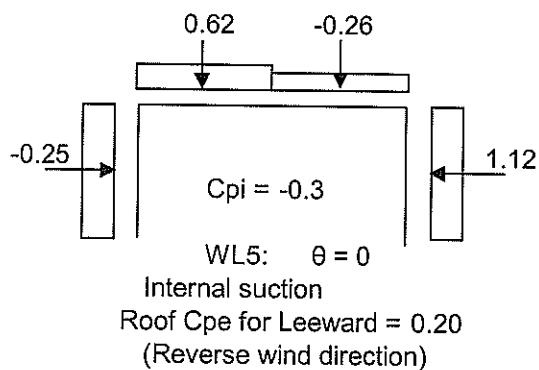
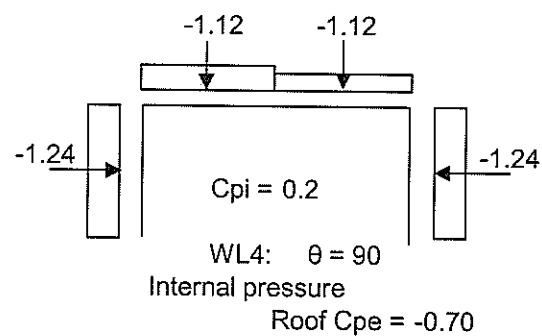
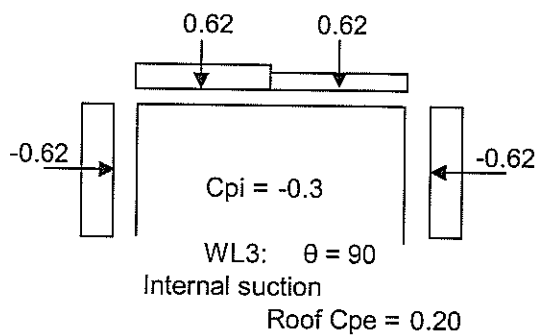
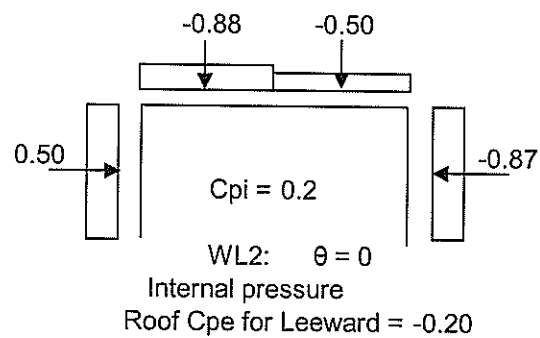
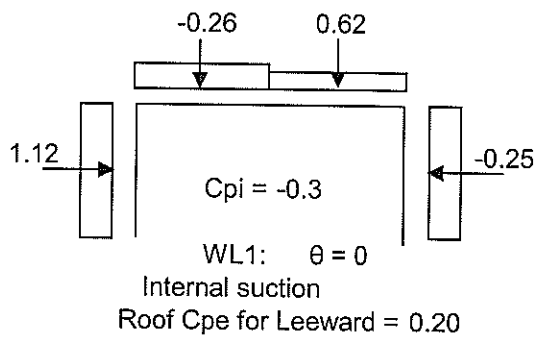
$$\begin{aligned}
 D &= 28.80 \text{ m} \\
 H &= 6.7 \text{ m} \\
 D/H &= 4.30
 \end{aligned}$$

Elements	Effective Wind Speed V_e m/s	Dynamic Wind Pressure q kN/m ²	External Pressure Coefficient, C_{pe}			
			$\theta = 0$ (normal to eave)		$\theta = 90$ (normal to gable end)	
			Windward	Leeward		
Roof	45	1.24	-0.51	-0.20 0.20	-0.7 (Zone C)	0.2 (Zone D)
Wall	45	1.24	0.60	-0.5 -0.5	-0.8 (Zone B)	-0.8 (Zone B)



Site:	Dubai Metro Project TPS Building Wind Load Computation	Date:	9-Jul-07	Job No.:	68010.01
		Designed by:	Rudy	Sheet No.:	

Wind Load = 1 x q x (Cpe - Cpi) kN/m (For unit width)





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Job Title

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Job No

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1

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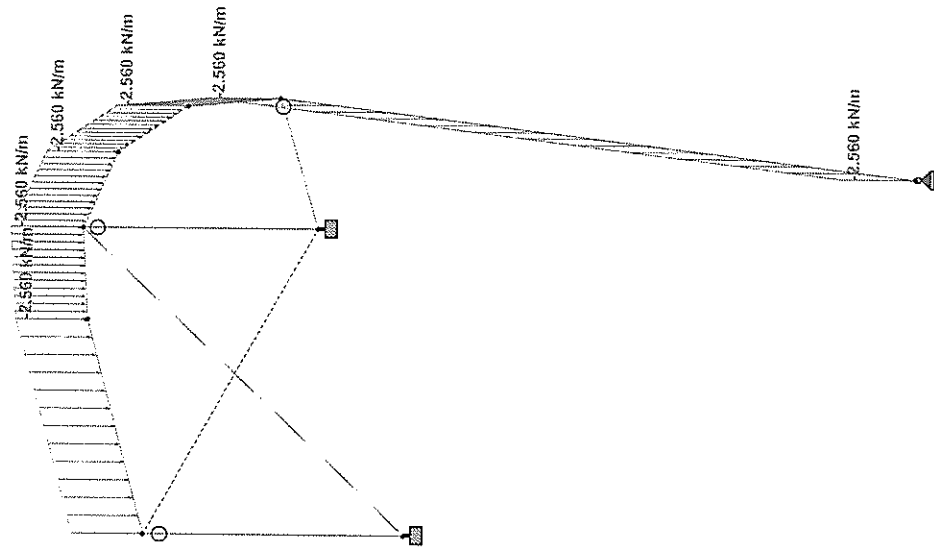
By

Date JULY 2007

Chd

File Steel_Roof-TPS.std

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1

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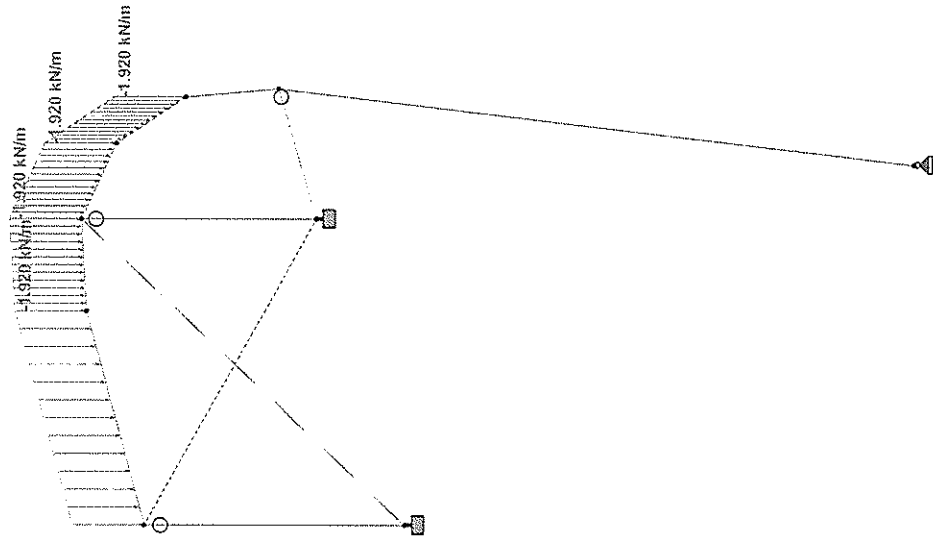
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Date JULY 2007

Chd

File Steel_Roof-TPS.std

Date/Time 09-Jul-2007 16:37



Load 3



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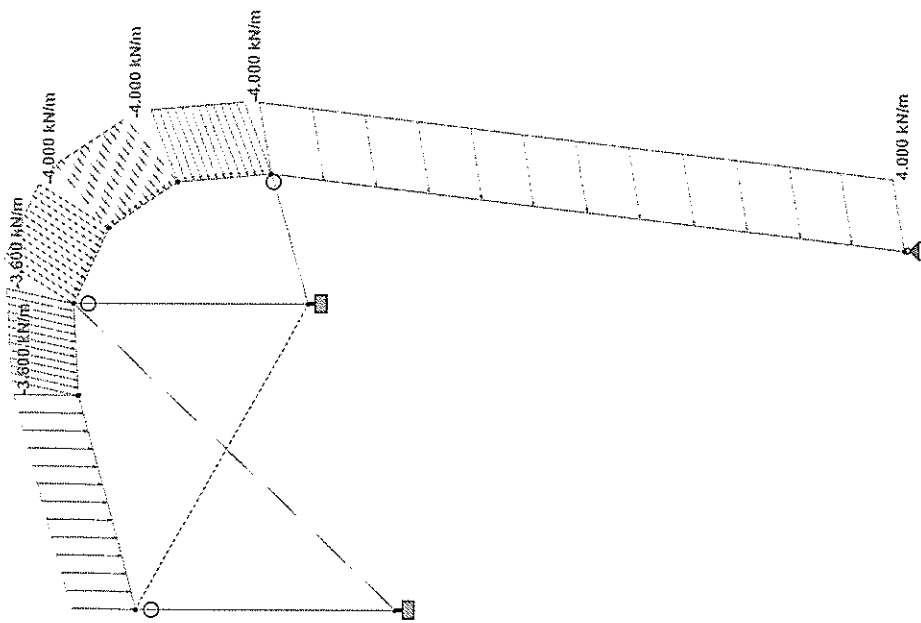
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By Date JULY 2007 Chd

File Steel_Roof-TPS.std Date/Time 09-Jul-2007 16:37

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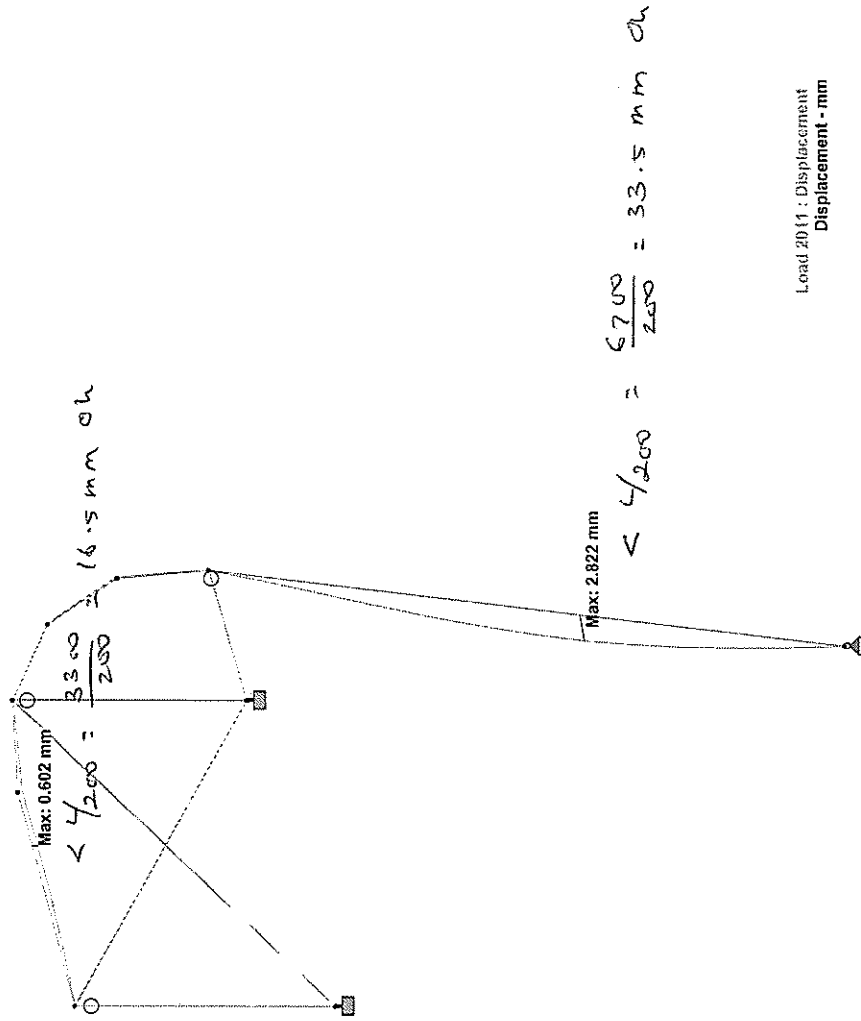


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Job No	Sheet No	Rev
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By	Date	Chd
	JULY 2007	
File Steel_Roof-TPS.std		
Date/Time 09-Jul-2007 16:37		



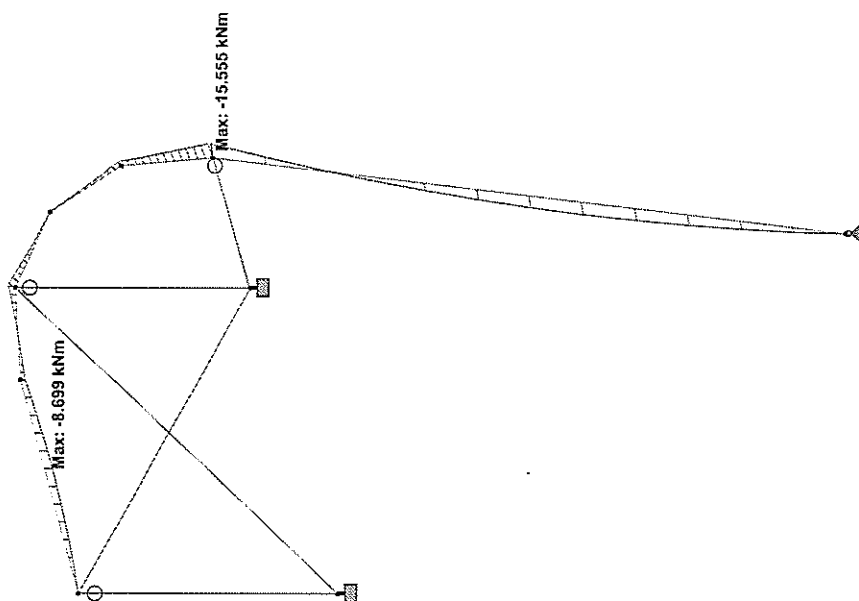


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Job Title

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Job No	Sheet No	Rev
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Part		
Ref		
By	Date	JULY 2007
		Chd
File	Steel_Roof-TPS.std	Date/Time 09-Jul-2007 16:37



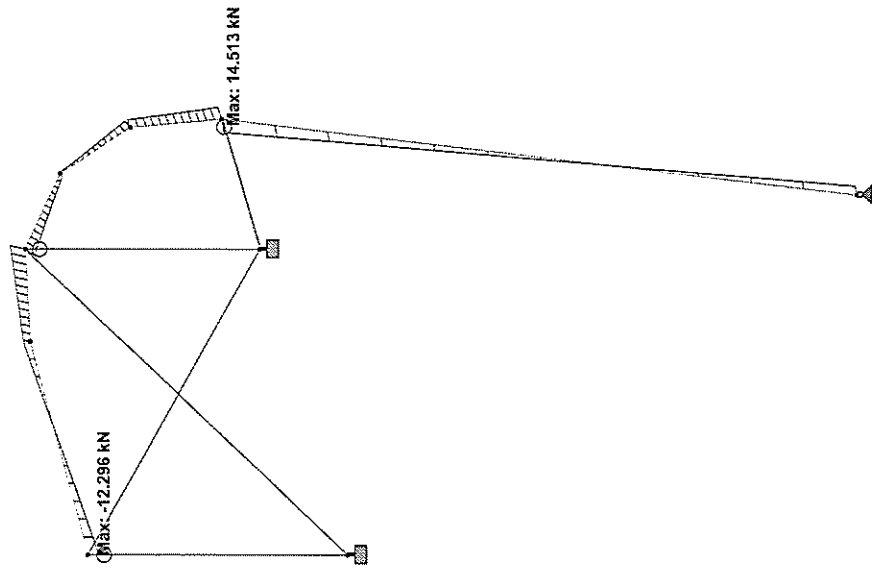


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	1	
Part		
Ref		
By	Date	Chd
	JULY 2007	
File	Steel_Roof-TPS.std	Date/Time
		09-Jul-2007 16:37



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*****
* STAAD.Pro
* Version 2006 Bld 1005.US
* Proprietary Program of
* Research Engineers, Intl.
* Date= JUL 9, 2007
* Time= 16:39:26
*
* USER ID: Meinhardt Infrastructure Pte Ltd
* *****

```

```

1. STAAD PLANE
INPUT FILE: Steel_Roof-TPS.STD
2. START JOB INFORMATION
3. ENGINEER DATE JULY 2007
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 1 10.50 0.00 0.00
9. 2 20.163 9.705
10. 3 20.163 9.705
11. 4 19.679 8.647
12. 5 18.850 9.272
13. 6 17.839 9.506
14. 7 15.501 9.547
15. 11 18.850 6.705
16. 12 15.501 6.705
18. MEMBER INCIDENCES
19. 1 1 2 6
20. 11 2 11
21. 12 11 5
22. 13 12 7
23. 14 13 5
24. 15 14 5
25. DEFINE MATERIAL START
26. ISOTROPIC STEEL
27. E 2.05E+08
28. E 2.05E+08
29. POISSON 0.3
30. DENSITY 76.8195
31. ALPHA 1.2E-005
32. DAMP 0.03
33. END DEFINE MATERIAL
35. MEMBER PROPERTY BRITISH
36. 1 TO 6 TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012
37. 11 TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012
38. 12 13 TABLE ST TUB1501506.3
39. 21 22 TABLE ST UA90X90X6
41. CONSTANTS
42. MATERIAL STEEL ALL
43. MEMBER TENSION
44. 21 22
45. MEMBER RELEASES
47. 11 START MX MY MZ
48. 12 13 END MX MY MZ
50. SUPPORTS
51. 1 PINNED
52. 11 12 FIXED
54. LOAD 1 SELFWEIGHT
55. SELFWEIGHT Y -1
57. LOAD 2 DEAD LOAD
58. MEMBER LOAD
59. *ASSUME ROOF M&E SERVICES LOAD = 0.3 KPA
60. *ASSUME ROOF INSULATION AND ROOF SHEETING AND MEMBERS LOAD = 0.5 KPA
61. 1 TO 6 UNIT GY -2.56
62. 11 TO 12 UNIT GY -2.56
63. 11 TO 12 UNIT GY -2.56
64. *UCL = 0.1 KPA
65. MEMBER LOAD
66. 3 TO 6 UNIT GY -1.92
68. LOAD 11 WL
69. MEMBER LOAD
70. 1 UNIT Y 4
71. 2 TO 4 UNIT Y -4.0
72. 5 6 UNIT Y -3.6
74. LOAD COMB 1001 1.4DL + 1.6LL
75. 1 1.4 2 1.4 3 1.6
76. LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL

```

-- PAGE NO. 2

```

77. 1 1.2 2 1.2 3 1.2 11 1.2
79. LOAD COMB 2001 1.0DL + 1.0LL
80. 1 1.0 2 1.0 3 1.0
81. LOAD COMB 2011 DL + LL + WL
82. 1 1.0 2 1.0 3 1.0 11 1.0
84. PERFORM ANALYSIS

```

PROBLEM STATISTICS

```

NUMBER OF JOINTS/MEMBERELEMENTS/SUPPORTS = 9/ 11/ 3
ORIGINAL/FINAL BANDWIDTH = 3
TOTAL DEGREES OF FREEDOM = 19
PRIMARY LOAD CASES = 4, 1 DOUBLE KILO-WORDS
SIZE OF STIFFNESS MATRIX = 12.0/ 366772.7 MB
REQD/AVAIL. DISK SPACE =

```

```

**START ITERATION NO. 2
**START ITERATION NO. 3
**NOTE-Tension/Compression converged after 3 iterations, Case= 1

```

```

**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 2
-- STAAD PLANE -- PAGE NO. 3

```

```

**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 3

```

```

**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 11

```

```

85. LOAD LIST ALL
86. PRINT SUPPORT REACTION
SUPPORT REACTION
STAAD PLANE

```

-- PAGE NO. 4

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = PLANE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
1	1	0.41	5.03	0.00	0.00	0.00	0.00
	2	1.59	20.18	0.00	0.00	0.00	0.00
	3	0.19	-4.79	0.00	0.00	0.00	0.00
1001	11	10.48	37.69	0.00	0.00	0.00	0.00
1011	15.21	26.30	0.00	0.00	0.00	0.00	0.00
2001	2.19	26.70	0.00	0.00	0.00	0.00	0.00
2011	12.67	21.92	0.00	0.00	0.00	0.00	0.00
11	1	-0.41	3.01	0.00	0.00	0.00	0.43
	2	-1.60	9.79	0.00	0.00	0.00	1.17
	3	-0.20	5.96	0.00	0.00	0.00	0.15
1001	11	29.67	15.50	0.00	0.00	0.00	-0.31
1011	32.14	28.71	0.00	0.00	0.00	0.00	1.73
2001	32.24	12.88	0.00	0.00	0.00	0.00	1.73
2011	27.45	32.95	0.00	0.00	0.00	0.00	1.44
12	1	0.00	2.05	0.00	0.00	0.00	-0.01
	2	0.01	3.94	0.00	0.00	0.00	-0.03
	3	-2.07	3.02	0.00	0.00	0.00	0.06
1001	0.03	13.21	0.00	0.00	0.00	0.00	-0.09
1011	-2.46	15.31	0.00	0.00	0.00	0.00	0.00
2001	0.02	9.01	0.00	0.00	0.00	0.00	-0.06
2011	-2.05	12.75	0.00	0.00	0.00	0.00	0.00

87. PRINT FORCE ENVELOPE NSECTION 5 ALL
 FORCE ENVELOPE NSECTION 5
 STAAD PLANE

-- PAGE NO. 5

MEMBER FORCE ENVELOPE

ALL UNITS ARE KN METE

MEMB	DISTANCE	FY	LD	FZ	LD	FZ	LD	MY	LD
1	0.00	MAX	1.51	1001	0.00	2011	0.00	2011	0.00
		MIN	-11.89	1011	0.00	2011	0.00	2011	0.00
	1.35	MAX	0.78	1001	0.00	2011	0.00	2011	0.00
		MIN	-6.03	1011	0.00	2011	0.00	2011	0.00
	2.70	MAX	0.05	1001	0.00	2011	0.00	2011	0.00
		MIN	-0.18	11	0.00	2011	0.00	2011	0.00
	4.05	MAX	5.69	1011	0.00	2011	0.00	2011	0.00
		MIN	-0.68	1001	0.00	2011	0.00	2011	0.00
	5.40	MAX	11.55	1011	0.00	2011	0.00	2011	0.00
		MIN	-1.41	1001	0.00	2011	0.00	2011	0.00
	6.76	MAX	17.42	1011	0.00	2011	0.00	2011	0.00
		MIN	-2.13	1001	0.00	2011	0.00	2011	0.00

MAX/MIN FORCE VALUES FOR MEMB

MEMB	DIST	FY	LD	FZ	LD	MY	LD	FX	LD
MAX.	17.42	6.76	1011	16.29	2.70	1011	0.00	37.78	C
MIN.	-11.89	0.00	1011	-18.67	6.76	1011	0.00	3.48	T

2	0.00	MAX	15.12	1011	18.67	1011	0.00	2011	0.00
		MIN	-4.06	1011	0.00	2011	0.00	2011	0.00
	0.21	MAX	12.99	1011	0.00	2011	0.00	2011	0.00
		MIN	-1.73	1011	0.00	2011	0.00	2011	0.00
	0.42	MAX	12.99	1011	0.00	2011	0.00	2011	0.00
		MIN	-1.73	1011	0.00	2011	0.00	2011	0.00
	0.62	MAX	11.93	1011	10.24	1011	0.00	2011	0.00
		MIN	-1.08	1001	-1.52	1011	0.00	2011	0.00
	0.83	MAX	10.86	1011	7.88	1011	0.00	2011	0.00
		MIN	-1.16	1001	-1.28	1011	0.00	2011	0.00
	1.04	MAX	9.80	1011	5.74	1011	0.00	2011	0.00
		MIN	-1.24	1001	-1.03	1001	0.00	2011	0.00

STAAD PLANE

MAX/MIN FORCE VALUES FOR MEMB

MEMB	DIST	FY	LD	FZ	LD	MY	LD	FX	LD
MAX.	15.12	0.00	1011	18.67	0.00	1011	0.00	9.09	C
MIN.	-1.24	1.04	1001	-2.12	0.00	1001	0.00	2.95	T

3	0.00	MAX	8.96	1011	5.74	1011	0.00	2011	0.00
		MIN	-0.00	2	-1.03	1011	0.00	2011	0.00
	0.21	MAX	7.35	1011	4.18	11	0.00	2011	0.00
		MIN	-0.26	1012	-1.40	1001	0.00	2011	0.00
	0.42	MAX	3.74	1011	1.37	1011	0.00	2011	0.00
		MIN	-4.84	11	-1.95	11	0.00	2011	0.00
	0.62	MAX	-1.56	1001	-0.77	1001	0.00	2011	0.00
		MIN	-3.71	11	-1.09	11	0.00	2011	0.00
	0.83	MAX	-2.32	1001	-0.37	2	0.00	2011	0.00
		MIN	-2.87	11	-0.61	1011	0.00	2011	0.00
	1.04	MAX	-3.07	1001	-0.13	2	0.00	2011	0.00

Page 3

Steel_Roof-TPS.ANL

3. AMONGST ALL SECT LOCATIONS

MAX/MIN	FORCE	VALUES	FOR	MEMB	FY	LD	FZ	LD	MY	LD	FX	LD
MAX.	8.96	0.00	1011	5.74	0.00	1011	0.00	0.00	1	4.63	C	0.00
MIN.	-3.07	1.04	1001	-1.10	0.21	1001	0.00	1.04	1011	9.49	T	1.04

4	0.00	MAX	0.04	11	0.61	1011	0.00	2011	0.00	2011	0.00	2011
		MIN	-3.89	1001	-0.13	2	0.00	1011	0.00	2011	0.00	2011
	0.21	MAX	-5.08	1011	0.00	2011	0.00	2011	0.00	2011	0.00	2011
		MIN	-0.58	1	0.17	1	0.00	2011	0.00	2011	0.00	2011
	0.42	MAX	-7.04	1011	4.37	1011	0.00	2011	0.00	2011	0.00	2011
		MIN	-0.68	1	0.30	1	0.00	2011	0.00	2011	0.00	2011
	0.62	MAX	-9.05	1011	6.45	1011	0.00	2011	0.00	2011	0.00	2011
		MIN	-0.78	1	0.45	1	0.00	2011	0.00	2011	0.00	2011
	0.83	MAX	-11.05	1011	8.96	1011	0.00	2011	0.00	2011	0.00	2011
		MIN	-0.88	1	0.62	1	0.00	2011	0.00	2011	0.00	2011
	1.04	MAX	-13.06	1011	0.00	2011	0.00	2011	0.00	2011	0.00	2011

STAAD PLANE

4. AMONGST ALL SECT LOCATIONS

MAX/MIN	FORCE	VALUES	FOR	MEMB	FY	LD	FZ	LD	MY	LD	FX	LD
MAX.	0.04	0.00	11	8.96	1.04	1011	0.00	0.00	1	0.07	C	0.00
MIN.	-13.06	1.04	1011	-0.13	0.00	2	0.00	1.04	1011	12.85	T	1.04

5	0.00	MAX	19.77	1011	8.96	1011	0.00	2011	0.00	2011	0.00	2011
		MIN	1.15	1	0.62	1	0.00	2011	0.00	2011	0.00	2011
	0.21	MAX	17.64	1011	5.08	1011	0.00	2011	0.00	2011	0.00	2011
		MIN	1.03	1	0.40	1	0.00	2011	0.00	2011	0.00	2011
	0.42	MAX	15.31	1011	1.96	1001	0.00	2011	0.00	2011	0.00	2011
		MIN	0.08	11	0.08	11	0.00	2011	0.00	2011	0.00	2011
	0.62	MAX	13.38	1011	1.36	1011	0.00	2011	0.00	2011	0.00	2011
		MIN	0.67	1	-0.13	1	0.00	2011	0.00	2011	0.00	2011
	0.83	MAX	11.25	1011	-3.92	1011	0.00	2011	0.00	2011	0.00	2011
		MIN	0.55	1	-0.26	1	0.00	2011	0.00	2011	0.00	2011
	1.04	MAX	9.12	1011	-6.03	1011	0.00	2011	0.00	2011	0.00	2011

5. AMONGST ALL SECT LOCATIONS

MAX/MIN	FORCE	VALUES	FOR	MEMB	FY	LD	FZ	LD	MY	LD	FX	LD
MAX.	19.77	0.00	1011	8.96	0.00	1011	0.00	0.00	1	4.96	C	0.00
MIN.	0.55	1.04	1011	-0.03	1.04	1011	0.00	1.04	1011	0.26	C	1.04

6	0.00	MAX	9.60	1011	-0.26	1	0.00	2011	0.00	2011	0.00	2011
		MIN	0.59	1	-6.03	1011	0.00	2011	0.00	2011	0.00	2011
	0.47	MAX	4.73	1011	-0.47	1	0.00	2011	0.00	2011	0.00	2011
		MIN	0.31	1	-9.38	1011	0.00	2011	0.00	2011	0.00	2011
	0.94	MAX	0.20	1001	-0.55	1	0.00	2011	0.00	2011	0.00	2011
		MIN	-0.25	1	-10.45	1011	0.00	2011	0.00	2011	0.00	2011
	1.40	MAX	-5.01	1011	-0.50	1	0.00	2011	0.00	2011	0.00	2011
		MIN	-0.33	1	-9.25	1011	0.00	2011	0.00	2011	0.00	2011
	1.87	MAX	-0.88	1011	-0.52	1011	0.00	2011	0.00	2011	0.00	2011
		MIN	-0.81	1	0.00	2011	0.00	2011	0.00	2011	0.00	2011
	2.34	MAX	-14.76	1011	0.00	2011	0.00	2011	0.00	2011	0.00	2011

STAAD PLANE

6. AMONGST ALL SECT LOCATIONS

MAX/MIN	FORCE	VALUES	FOR	MEMB	FY	LD	FZ	LD	MY	LD	FX	LD
---------	-------	--------	-----	------	----	----	----	----	----	----	----	----

Page 4

Steel_Roof-TPS.AHL

MAX.	9.60	0.00	10.01	0.00	2.34	1	1.74	C	0.00	10.01
MIN.	-14.76	2.34	10.01	-10.45	0.94	10.01	0.11	T	2.34	11
	0.00	2.34	20.01	0.00	0.94	10.01	0.11	T	2.34	11

11	0.00	MAX	0.30	11	0.00	20.01	0.00	20.01	0.00	20.01
	0.28	MIN	-1.18	10.01	0.00	20.01	0.00	20.01	0.00	20.01
	0.57	MAX	0.30	11	0.37	10.01	0.00	20.01	0.00	20.01
	0.85	MIN	-1.42	10.01	-0.09	11	0.00	20.01	0.00	20.01
	1.14	MAX	0.30	11	0.43	10.01	0.00	20.01	0.00	20.01
	1.42	MIN	-1.89	10.01	-0.36	11	0.00	20.01	0.00	20.01
		MAX	0.30	11	0.36	11	0.00	20.01	0.00	20.01
		MIN	-1.89	10.01	-0.36	11	0.00	20.01	0.00	20.01
		MAX	0.30	11	0.36	11	0.00	20.01	0.00	20.01
		MIN	-1.89	10.01	-0.36	11	0.00	20.01	0.00	20.01
		MAX	0.30	11	0.36	11	0.00	20.01	0.00	20.01
		MIN	-1.89	10.01	-0.36	11	0.00	20.01	0.00	20.01

MAX/MIN FORCE VALUES FOR MEMB

	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.30	0.00	11	2.52	1.42	1001			
	0.00	0.00	1	0.00	0.00	1	31.70	C	0.00 1011
	-2.37	1.42	1001	-0.43	1.42	11			
MIN.	0.00	1.42	2011	1.42	1.42	2011	4.73	T	1.42 1001

12	0.00	MAX	0.05	11	0.12	11	0.00	20.01	0.00	20.01
	0.51	MIN	-0.02	10.01	-0.04	10.01	0.00	20.01	0.00	20.01
	1.03	MAX	0.05	11	0.10	11	0.00	20.01	0.00	20.01
	1.54	MIN	-0.02	10.01	-0.03	10.01	0.00	20.01	0.00	20.01
	2.05	MAX	0.05	11	0.07	11	0.00	20.01	0.00	20.01
	2.57	MIN	-0.02	10.01	-0.02	10.01	0.00	20.01	0.00	20.01
		MAX	0.05	11	0.05	11	0.00	20.01	0.00	20.01
		MIN	-0.02	10.01	-0.02	10.01	0.00	20.01	0.00	20.01
		MAX	0.05	11	0.02	11	0.00	20.01	0.00	20.01
		MIN	-0.02	10.01	-0.01	10.01	0.00	20.01	0.00	20.01
		MAX	0.05	11	0.00	20.01	0.00	20.01	0.00	20.01
		MIN	-0.02	10.01	0.00	20.01	0.00	20.01	0.00	20.01

MAX/MIN FORCE VALUES FOR MEMB

POORLY PLANNED TOURISM DEVELOPMENT				POORLY PLANNED TOURISM DEVELOPMENT				POORLY PLANNED TOURISM DEVELOPMENT			
FY/	FZ/	DIST	LD	FY/	FZ/	DIST	LD	FY/	FZ/	DIST	LD
0.05	0.00	0.00	11	0.12	0.00	0.00	11	0.12	0.00	0.00	11
0.00	0.00	0.00	11	0.00	0.00	0.00	11	0.00	0.00	0.00	11
-0.02	2.57	20.01		-0.04	0.00	0.00	10.01	-0.04	0.00	0.00	10.01
0.00	2.57	20.01		0.00	2.57	20.01		0.00	2.57	20.01	
		</									

13	0.00	MAX	0.02	11	0.06	11	0.00	20.01	0.00	20.01
	0.57	MIN	-0.03	10.01	-0.09	10.01	0.00	20.01	0.00	20.01
	1.14	MAX	0.02	11	0.05	11	0.00	20.01	0.00	20.01
	1.71	MIN	-0.03	10.01	-0.07	10.01	0.00	20.01	0.00	20.01
	2.27	MAX	0.02	11	0.04	11	0.00	20.01	0.00	20.01
	2.84	MIN	-0.03	10.01	-0.05	10.01	0.00	20.01	0.00	20.01
		MAX	0.02	11	0.02	11	0.00	20.01	0.00	20.01
		MIN	-0.03	10.01	-0.04	10.01	0.00	20.01	0.00	20.01
		MAX	0.02	11	0.01	11	0.00	20.01	0.00	20.01
		MIN	-0.03	10.01	-0.02	10.01	0.00	20.01	0.00	20.01
		MAX	0.02	11	0.00	20.01	0.00	20.01	0.00	20.01
		MIN	-0.03	10.01	0.00	20.01	0.00	20.01	0.00	20.01

MAX/MIN FORCE VALUES FOR MEMB

MAX/MIN FORCE VALUES FOR MEMS				ALL TEST LOCATIONS			
FY/	DIST	LD		MZ/	DIST	LD	
FZ/	DIST	LD		NY/	DIST	LD	
MAX,	0.02	0.00	11	0.06	0.00	11	
	0.00	0.00	1	0.00	0.00	1	16.98 C
	0.00	0.00	1	-0.09	0.00	1001	
MIN,	-0.03	2.84	1001				1.09 C
	0.00	2.84	2011		2.84	2011	2.84 1

Page 5

Steel_Roof-TPS.AHL

21	0.00	MAX	0.00	20.01	0.00	20.01	0.00	20.01	0.00	20.01
	0.84	MIN	0.00	20.01	0.00	20.01	0.00	20.01	0.00	20.01
	1.69	MAX	0.00	20.01	0.00	20.01	0.00	20.01	0.00	20.01
	2.53	MIN	0.00	20.01	0.00	20.01	0.00	20.01	0.00	20.01
	3.38	MAX	0.00	20.01	0.00	20.01	0.00	20.01	0.00	20.01
	4.22	MIN	0.00	20.01	0.00	20.01	0.00	20.01	0.00	20.01
		MAX	0.00	20.01	0.00	20.01	0.00	20.01	0.00	20.01
		MIN	0.00	20.01	0.00	20.01	0.00	20.01	0.00	20.01

STAAD PLANE

MAX/MIN FORCE VALUES FOR MEMB

	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.00	0.00	1	0.00	0.00	1			
	0.00	0.00	1	0.00	0.00	1	0.00	0.00	1
	0.00	4.22	2011	0.00	4.22	2011			
MIN.	0.00	4.22	2011	0.00	4.22	2011	3.10	T	4.22 1011

22	0.00	MAX	0.19	10.01	0.00	20.01	0.00	20.01	0.00	20.01
	0.88	MIN	0.00	20.01	0.00	20.01	0.00	20.01	0.00	20.01
	1.76	MAX	0.00	20.01	0.00	20.01	0.00	20.01	0.00	20.01
	2.64	MIN	0.00	20.01	0.00	20.01	0.00	20.01	0.00	20.01
	3.51	MAX	0.00	20.01	0.00	20.01	0.00	20.01	0.00	20.01
	4.39	MIN	0.00	20.01	0.00	20.01	0.00	20.01	0.00	20.01
		MAX	0.00	20.01	0.00	20.01	0.00	20.01	0.00	20.01
		MIN	-0.19	10.01	0.00	20.01	0.00	20.01	0.00	20.01

MAX/MIN FORCE VALUES FOR MEMB

	FY/	DIST	LD	MZ/	DIST	LD	FX	DIST	LD
MAX.	0.19	0.00	1001	0.00	0.00	1			
	0.00	0.00	1	0.00	0.00	1	0.00	0.00	11
MIN.	-0.19	4.39	1001	0.00	4.39	1011			
	0.00	4.39	1011	0.00	4.39	1011	2.23	T	0.00 1001

***** END OF FORCE ENVELOPE FROM INTERNAL STORAGE *****

88. LOAD LIST 2001 2011

89. PRINT JOINT DISPL

STAAD PLANE

--- PAGE NO. 11

JOINT DISPLACEMENT (CM RADIANS)

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
1	2001	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0002
2	2001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0015
3	2001	-0.0057	-0.0071	0.0000	0.0000	0.0000	-0.0001
4	2001	-0.0184	-0.0074	0.0000	0.0000	0.0000	0.0000
5	2001	-0.0039	-0.0074	0.0000	0.0000	0.0000	0.0000
6	2001	-0.0134	-0.0058	0.0000	0.0000	0.0000	0.0001
7	2001	-0.0075	-0.0064	0.0000	0.0000	0.0000	0.0002
8	2001	-0.0080	-0.0119	0.0000	0.0000	0.0000	0.0002
9	2001	-0.0071	-0.0270	0.0000	0.0000	0.0000	0.0001
10	2001	-0.0068	-0.0512	0.0000	0.0000	0.0000	0.0003
11	2001	-0.0065	-0.0533	0.0000	0.0000	0.0000	-0.0003

Page 6

2011	0.0001	-0.0053	0.0000	0.0000	0.0000	-0.0005
11 2001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
12 2001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2011	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

***** END OF LATEST ANALYSIS RESULT *****

90. PARAMETER 1
 91. CODE BS5950
 92. TRACK 2 ALL
 93. TRACK 2 ALL
 94. CHECK CODE ALL
 STEEL DESIGN

STAAD.PTO CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.9_5950-1_2000

STAAD PLANE

--- PAGE NO. 12

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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1 TAPERED PASS 14.47 C BS-4.7 (C) 0.00 0.061 2011

13.57	2.81
CALCULATED CAPACITIES FOR MEMB 1 UNIT - KN/m SECTION CLASS 3	
MCZ= 270.6	MCY= 111.6 PC= 1048.0 PT= 0.0 MB= 165.0
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00	
Shear Buckling check is required: Vb = 332 kN : qw = 207 N/mm2	
d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS	

2 TAPERED PASS 3.30 C BS-4.7 (C) 0.00 0.059 2011

13.56	0.00
CALCULATED CAPACITIES FOR MEMB 2 UNIT - KN/m SECTION CLASS 3	
MCZ= 270.6	MCY= 111.6 PC= 2677.2 PT= 0.0 MB= 270.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00	
Shear Buckling check is required: Vb = 332 kN : qw = 207 N/mm2	
d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS	

3 TAPERED PASS 3.29 T BS-4.8.2.2 0.00 0.019 2011

4.78	0.00
CALCULATED CAPACITIES FOR MEMB 3 UNIT - KN/m SECTION CLASS 3	
MCZ= 270.6	MCY= 111.6 PC= 2677.1 PT= 2693.8 MB= 270.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00	
Shear Buckling check is required: Vb = 332 kN : qw = 207 N/mm2	
d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS	

4 TAPERED PASS 10.71 T BS-4.8.2.2 0.00 0.032 2011

-7.46	1.04
CALCULATED CAPACITIES FOR MEMB 4 UNIT - KN/m SECTION CLASS 3	
MCZ= 270.6	MCY= 111.6 PC= 11908.2 PT= 2693.8 MB= 270.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00	
Shear Buckling check is required: Vb = 332 kN : qw = 207 N/mm2	
d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS	

5 TAPERED PASS 3.90 C BS-4.7.3-(Y) 0.00 0.040 2011

7.46	0.00
STAAD PLANE	

--- PAGE NO. 13

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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Page 7

CALCULATED CAPACITIES FOR MEMB 5 UNIT - KN/m SECTION CLASS 3	
MCZ= 270.6	MCY= 111.6 PC= 2677.3 PT= 0.0 MB= 270.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00	
Shear Buckling check is required: Vb = 332 kN : qw = 207 N/mm2	
d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS	

6 TAPERED PASS 0.98 C BS-4.7 (C) 0.00 -8.70 2011

-0.97	0.00
CALCULATED CAPACITIES FOR MEMB 6 UNIT - KN/m SECTION CLASS 3	
MCZ= 270.6	MCY= 111.6 PC= 2346.9 PT= 0.0 MB= 270.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00	
Shear Buckling check is required: Vb = 332 kN : qw = 207 N/mm2	
d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS	

11 TAPERED PASS 26.41 C BS-4.7 (C) 0.00 -1.35 2011

-1.42	0.015
CALCULATED CAPACITIES FOR MEMB 11 UNIT - KN/m SECTION CLASS 3	
MCZ= 270.6	MCY= 111.6 PC= 2583.3 PT= 2693.8 MB= 270.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00	
Shear Buckling check is required: Vb = 332 kN : qw = 207 N/mm2	
d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS	

STAAD PLANE

--- PAGE NO. 14

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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12 ST TUB1501506.3 PASS BS-4.8.3.3 0.00 0.032 2011

0.09	2011
MATERIAL DATA	
Grade of steel = S 275	
Modulus of elasticity = 205 kN/mm2	
Design strength (py) = 344 N/mm2	
SECTION PROPERTIES (units - cm)	
Member Length = 256.70	
Gross Area = 35.80 Net Area = 35.80 Eff. Area = 35.80	

Moment of inertia	
Plastic modulus	
Elastic modulus	
Effective modulus	
Shear Area	
DESIGN DATA (units - kN m)	
Section Class	
Squash Load	
Axial force/Squash load	
z-z axis	
y-y axis	

Compression Capacity	
Moment Capacity	
Reduced Moment Capacity	
Shear Capacity	
BUCKLING CALCULATIONS (units - kN,m)	
(axis nomenclature as per design code)	
Slenderness	
Radius of gyration (cm)	
Effective length	
LTA check unnecessary for this section	

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):	
CLAUSE	
BS-4.7 (C)	
BS-4.8.3.2	

FX	VY	VZ	MZ	MY
0.030	34.5	-	-	-
0.028	2011	0.0	0.0	0.0

Page 8

85-4.8.3.3.1 0.032 2011 34.5 - 0.1 0.0
 BS-4.8.3.3.3 0.032 2011 34.5 - 0.1 0.0
 ANNEX I.1 0.001 2011 34.5 - 0.1 0.0
 Torsion and deflections have not been considered in the design.

-- PAGE NO. 15

STAAD PLANE

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/	CRITICAL COND/	RATIO/	LOADING/
		FX	MY	MZ	LOCATION

13 ST TUB1501506.3 PASS 85-4.8.3.3.3 0.013 2011
 14.15 C 0.00 0.00 --

MATERIAL DATA

Grade of steel = S 275
 Modulus of elasticity = 205 kN/mm2
 Design strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 284.20
 Gross Area = 35.80 Net Area = 35.80 Eff. Area = 35.80

Moment of inertia : 1223.000 Y-Y axis
 Plastic modulus : 192.000
 Elastic modulus : 163.067
 Effective modulus : 163.067
 Shear Area : 17.900

DESIGN DATA (units - kN,m)

Section Class : SEMI-COMPACT
 Squash Load : 1235.10
 Axial force/Squash Load : 0.011

Compression Capacity : 1117.9 Y-Y axis
 Moment Capacity : 66.2
 Reduced Moment Capacity : 66.2
 Shear Capacity : 370.5

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)
 X-X axis
 Slenderness : 48.624
 Radius of gyration (cm) : 5.845
 Effective Length : 2.842

LTB check unnecessary for this section

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m) :

CLAUSE	RATIO	LOAD	FX	VY	VZ	MY
BS-4.7 (C)	0.013	2011	14.2	0.0	0.0	0.0
BS-4.8.3.2	0.011	2011	14.2	0.0	0.0	0.0
BS-4.8.3.3.1	0.013	2011	14.2	0.0	0.0	0.0
BS-4.8.3.3.3	0.013	2011	14.2	0.0	0.0	0.0

Torsion and deflections have not been considered in the design.

-- PAGE NO. 16

STAAD PLANE

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/	CRITICAL COND/	RATIO/	LOADING/
		FX	MY	MZ	LOCATION

21 ST UA90X90X6 PASS 85-4.6 (T) 0.007 2011
 2.58 T 0.00 0.00 --

MATERIAL DATA

Grade of steel = S 275
 Modulus of elasticity = 205 kN/mm2
 Design strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 284.20
 Gross Area = 35.80 Net Area = 35.80 Eff. Area = 35.80

Moment of inertia : 1223.000 Y-Y axis
 Plastic modulus : 192.000
 Elastic modulus : 163.067
 Effective modulus : 163.067
 Shear Area : 17.900

DESIGN DATA (units - kN,m)

Section Class : SEMI-COMPACT
 Squash Load : 1235.10
 Axial force/Squash Load : 0.011

Compression Capacity : 1117.9 Y-Y axis
 Moment Capacity : 66.2
 Reduced Moment Capacity : 66.2
 Shear Capacity : 370.5

Member Length = 421.96 Steel_Roof-TPS.ANL
 Gross Area = 10.60 Net Area = 10.60 Eff. Area = 10.60

-- PAGE NO. 17

STAAD PLANE

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/	CRITICAL COND/	RATIO/	LOADING/
		FX	MY	MZ	LOCATION

22 ST UA90X90X6 PASS 85-4.6 (T) 0.004 2001
 1.53 T 0.00 0.00 --

MATERIAL DATA

Grade of steel = S 275
 Modulus of elasticity = 205 kN/mm2
 Design strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 439.24
 Gross Area = 10.60 Net Area = 10.60 Eff. Area = 10.60

Moment of inertia : 131.618 Y-Y axis
 Plastic modulus : 17.554
 Elastic modulus : 9.752
 Effective modulus : 9.752
 Shear Area : 3.240

DESIGN DATA (units - kN,m)

Section Class : SLENDER
 Tension Capacity : 365.7
 Shear Capacity : 67.1

BUCKLING CALCULATIONS (units - kN,m)
 (axis nomenclature as per design code)
 LTB Moment Capacity (kNm) and LTB Length (m): 3.99, 4.392
 LTB Coefficients & Associated Moments (kNm):
 mlt = 1.00 : mx = 1.00 : my = 1.00 : myx = 1.00
 Mit = 0.00 : mx = 0.00 : my = 0.00 : my = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m) :

CLAUSE	RATIO	LOAD	FX	VY	VZ	MY
BS-4.6 (T)	0.007	2011	2.6	0.0	0.0	0.0

Torsion and deflections have not been considered in the design.

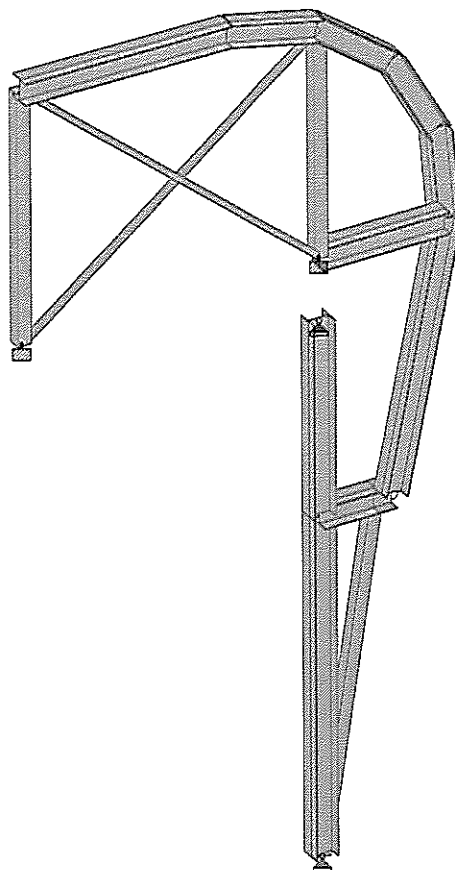
***** END OF TABULATED RESULT OF DESIGN *****

Page 10



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Job No	Sheet No 1	Rev
Part		
Ref		
By	Date JULY 2007	Chd
Client	File BRU Steel_Roof-TPS.std	Date/Time 18-Jul-2007 15:54



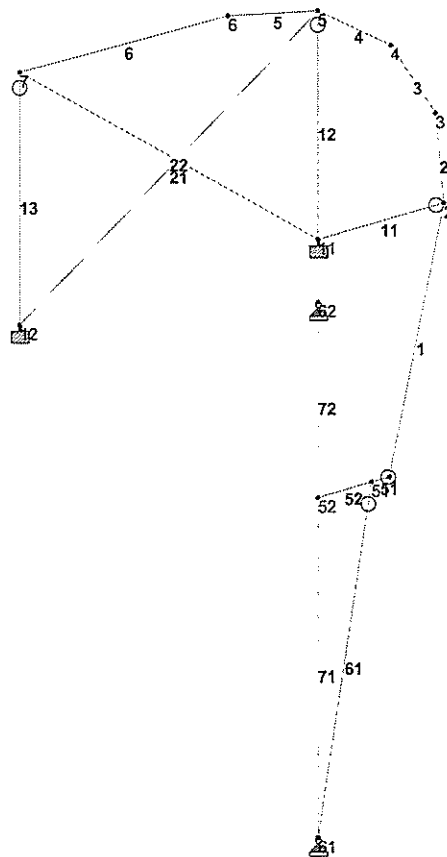
Load 1

SIDE FRAME-BRU PANEL AREA (TPS)



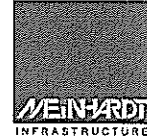
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Job No	Sheet No 1	Rev
Part		
Ref		
By	Date JULY 2007	Chd
Client	File BRU Steel_Roof-TPS.std	Date/Time 18-Jul-2007 15:54

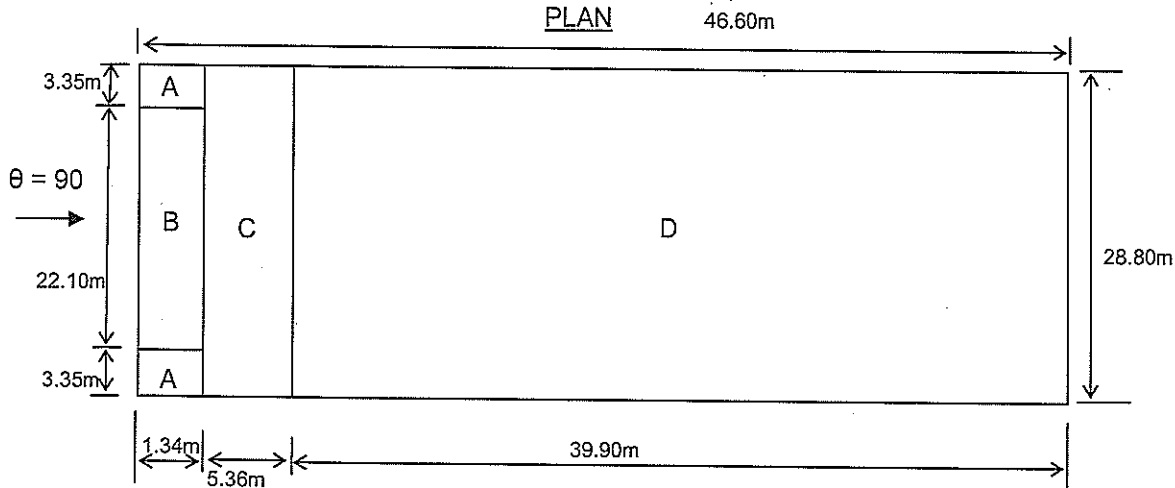


Load 1

JOINT & MEMBER INCIDENCES



Site:	Dubai Metro Project	Date:	9-Jul-07	Job No.:	68010.01
	TPS Building				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to gable end)


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 28.8 \text{ m} \\
 H &= 6.7 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 2 \times 6.7 \\
 &= 13.4 \text{ m}
 \end{aligned}$$

For maximum uplift, assume zone C coefficient =>

$$C_{pe} = -0.7$$

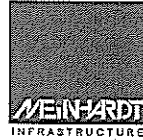
For maximum downward force, assume zone D coefficient =>

$$C_{pe} = 0.2$$

Wind Load Computation

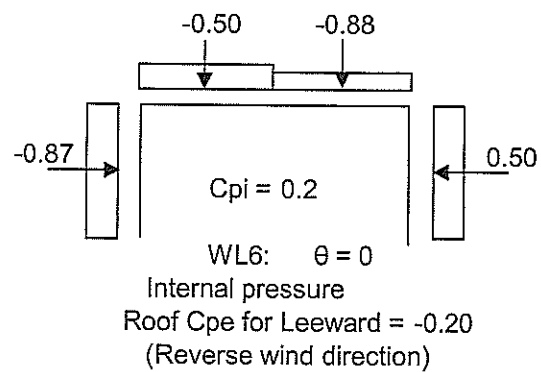
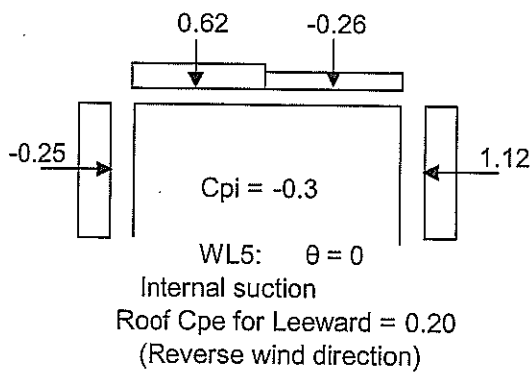
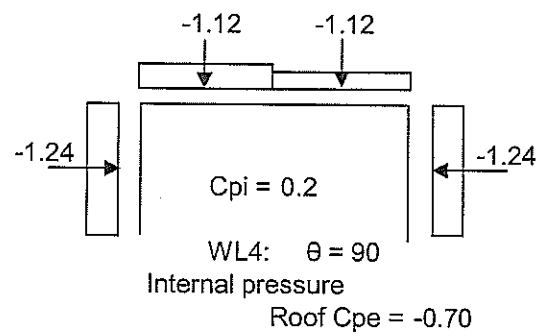
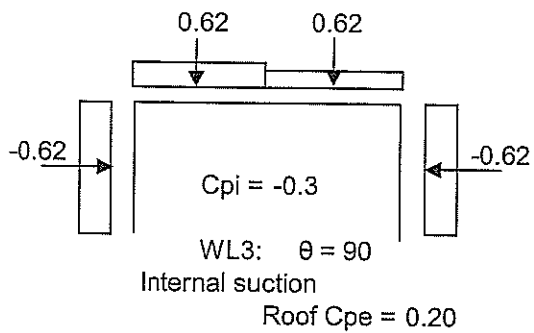
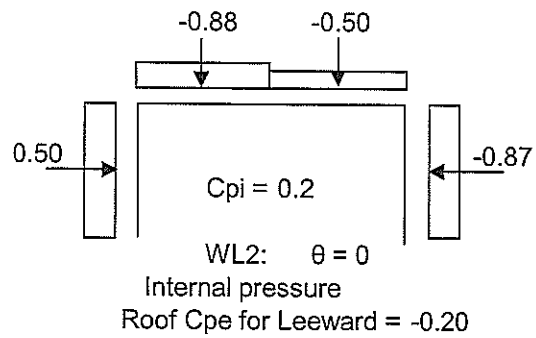
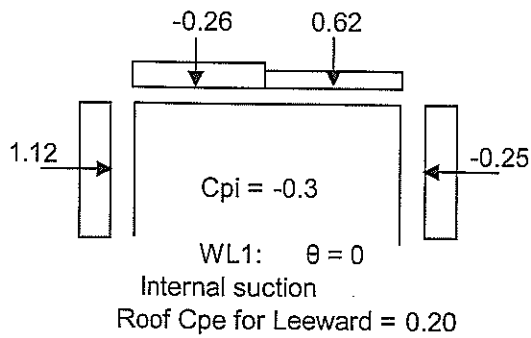
$$\begin{aligned}
 D &= 28.80 \text{ m} \\
 H &= 6.7 \text{ m} \\
 D/H &= 4.30
 \end{aligned}$$

Elements	Effective Wind Speed V_e m/s	Dynamic Wind Pressure q kN/m ²	External Pressure Coefficient, C_{pe}			
			$\theta = 0$ (normal to eave)		$\theta = 90$ (normal to gable end)	
			Windward	Leeward		
Roof	45	1.24	-0.51	-0.20 0.20	-0.7 (Zone C)	0.2 (Zone D)
Wall	45	1.24	0.60	-0.5 -0.5	-0.8 (Zone B)	-0.8 (Zone B)



Site:	Dubai Metro Project TPS Building Wind Load Computation	Date:	9-Jul-07	Job No.:	68010.01
		Designed by:	Rudy	Sheet No.:	

Wind Load = 1 x q x (Cpe - Cpi) kN/m (For unit width)





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Job No

Sheet No

1

Rev

Part

Job Title

Ref

By

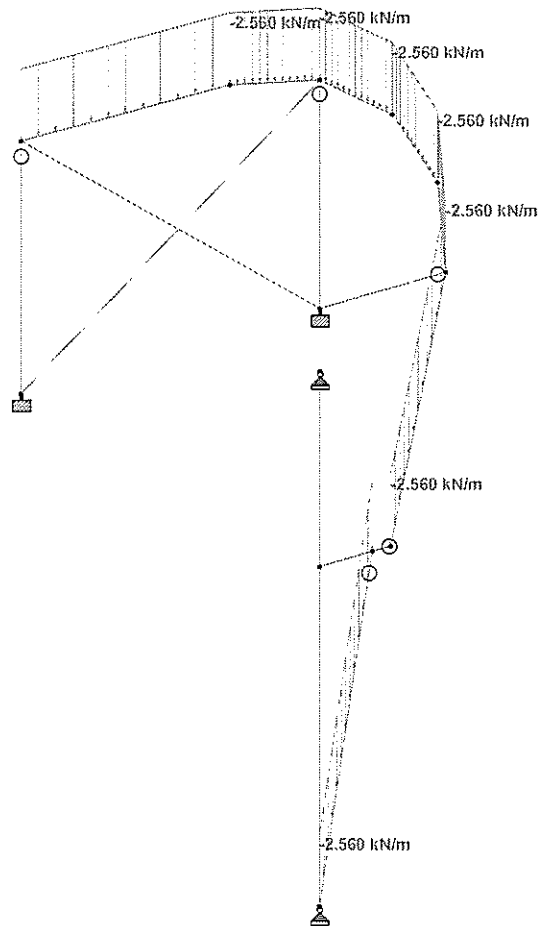
Date JULY 2007

Chd

Client

File BRU Steel_Roof-TPS.std

Date/Time 18-Jul-2007 15:54

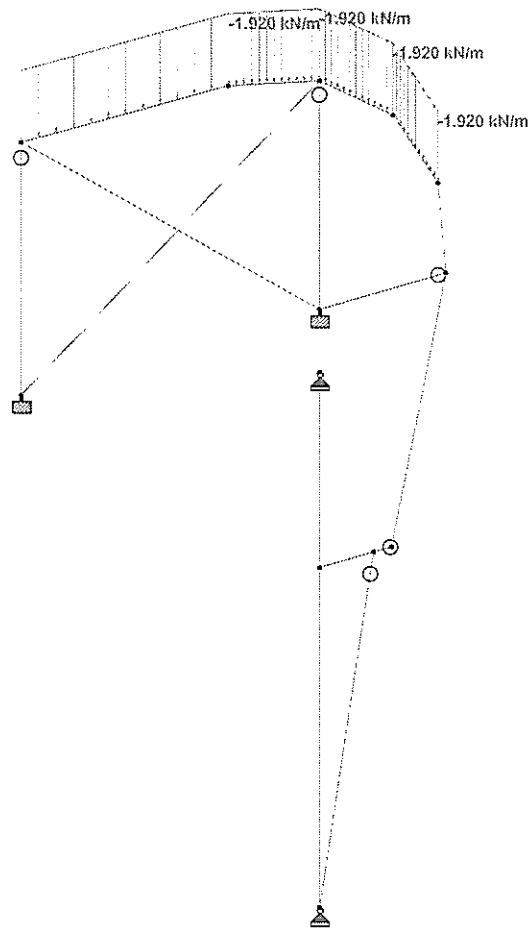


Load 2



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Job No	Sheet No 1	Rev
Part		
Ref		
By	Date JULY 2007	Chd
Client	File BRU Steel_Roof-TPS.std	Date/Time 18-Jul-2007 15:54



Load 3



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Job No

Sheet No

1

Rev

Part

Job Title

Ref

By

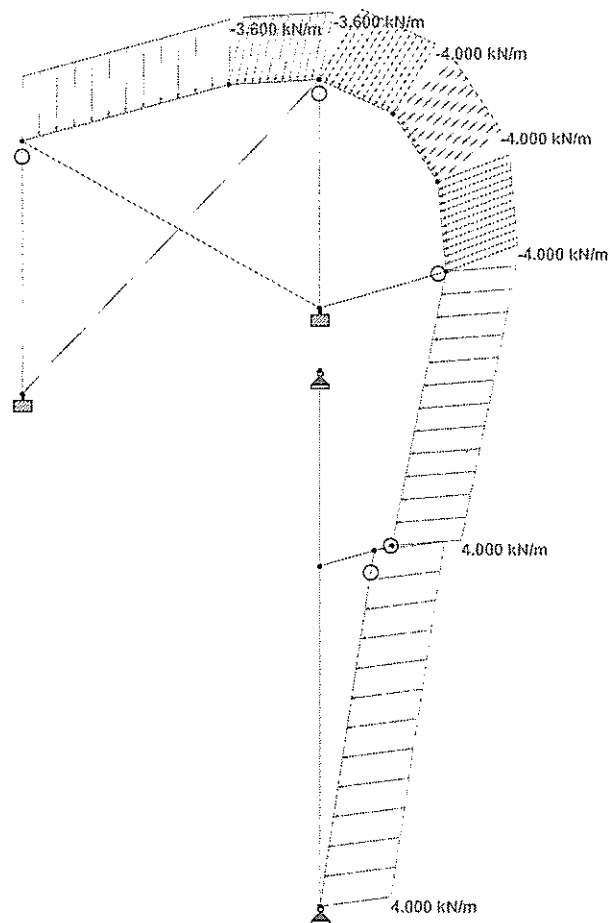
Date JULY 2007

Chd

Client

File BRU Steel_Roof-TPS.std

Date/Time 18-Jul-2007 15:54

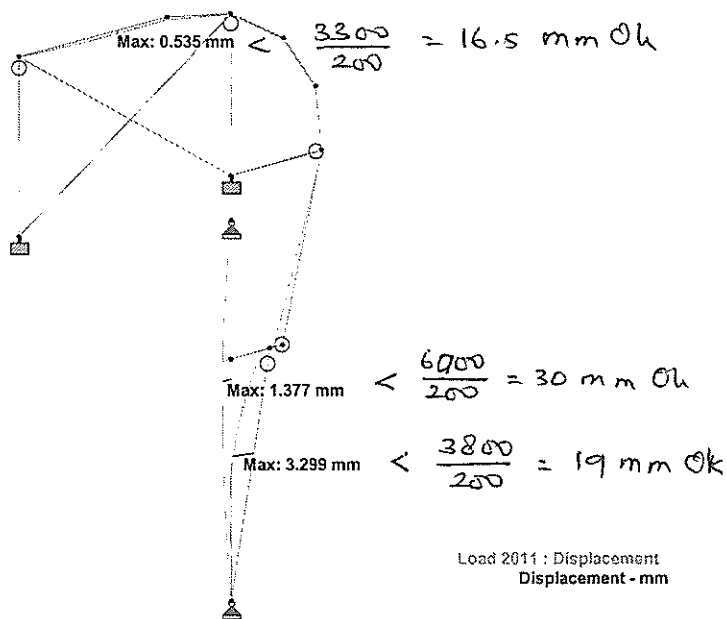


Load 11



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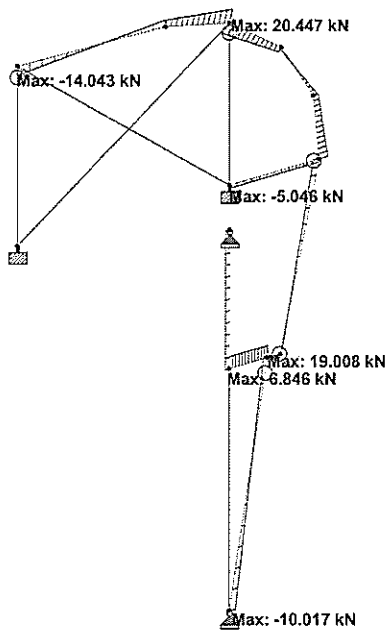
Job No	Sheet No 1	Rev
Part		
Ref		
By	Date JULY 2007	Chd
Client	File BRU Steel_Roof-TPS.std	Date/Time 18-Jul-2007 15:54





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Job No	Sheet No 1	Rev
Part		
Ref		
By	Date JULY 2007	Chd
Client	File BRU Steel_Roof-TPS.std	Date/Time 18-Jul-2007 15:54



Load 1011 : Shear Y
Force - kN



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Job No

Sheet No

1

Rev

Part

Job Title

Ref

By

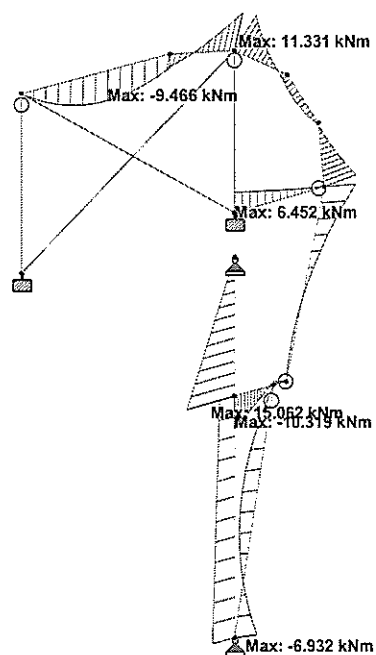
Date JULY 2007

Chd

Client

File BRU Steel_Roof-TPS.std

Date/Time 18-Jul-2007 15:54



Load 1011 : Bending Z
Moment - kNm

* STAAD.Pro
* Version 2006 Bld 1005.US
* Proprietary Program of
* Research Engineers, Intl.
* Date= JUL 18, 2007
* Time= 15:59:43
*
* USER ID: Weinhardt Infrastructure Pte Ltd
* *****

1. STAAD PLANE
INPUT FILE: BRU Steel1_Roof-TPS.STD

2. END JOB INFORMATION
3. ENGINEER DATE JULY 2007
4. END JOB INFORMATION

5. INPUT WIDTH 79

6. UNIT METER KN

7. JOINT COORDINATES

8. 1 19.656 3.800

9. 2 20.273 6.705

10. 3 20.183 7.739

11. 4 19.679 8.647

12. 5 18.850 9.272

13. 6 17.839 9.506

14. 7 15.501 9.547

15. 11 18.850 6.705

16. 12 15.301 6.705

17. 13 18.850 9.272

18. 14 18.850 9.506

19. 18 18.850 3.800

20. 62 18.850 6.000

21. 62 18.850 6.000

22. MEMBER INCIDENCES

23. 1 1 2 6

24. 11 2 11

25. 12 11 5

26. 13 12 7

27. 21 12 5

28. 22 7 11

29. 51 1 51

30. 52 51 52

31. 61 61 51

32. 71 61 52

33. 72 52 62

34. DEFINE MATERIAL START

35. 1 STEEL

36. E 210000

37. G 0.00003

38. POISSON 0.3

39. DENSITY 76.8195

40. ALPHA 1.2E-005

41. DAMP 0.03

42. END DEFINE MATERIAL

43. STAAD PLANE

44. MEMBER PROPERTY BRITISH

45. 1 TO 6 TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012

46. 11 TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012

47. 12 13 TABLE ST TUB1501506.3

48. 21 22 TABLE ST TUB1501506.3

49. 31 32 TABLE ST TUB1501506.3

50. 61 62 TABLE ST TUB1501506.3

51. 71 72 TAPERED 0.25 0.008 0.25 0.25 0.012 0.25 0.012

52. CONSTANTS

53. MATERIAL STEEL ALL

54. MEMBER TENSION

55. 21 22

56. MEMBER RELEASES

57. 11 51 START MX MY MZ

58. 12 13 61 END MX MY MZ

59. SUPPORTS

60. 61 62 PINNED

61. LOAD 1 SELFWEIGHT

62. LOAD 2 DEAD LOAD

63. SELFWEIGHT Y -1

64. MEMBER RELEASES

65. 11 51 START MX MY MZ

66. 12 13 61 END MX MY MZ

67. 21 22 31 32 41 42 51 52 61 62 71 72 81 82 91 92 101 102 111 112 121 122 131 132 141 142 151 152 161 162 171 172 181 182 191 192 201 202 211 212 221 222 231 232 241 242 251 252 261 262 271 272 281 282 291 292 301 302 311 312 321 322 331 332 341 342 351 352 361 362 371 372 381 382 391 392 401 402 411 412 421 422 431 432 441 442 451 452 461 462 471 472 481 482 491 492 501 502 511 512 521 522 531 532 541 542 551 552 561 562 571 572 581 582 591 592 601 602 611 612 621 622 631 632 641 642 651 652 661 662 671 672 681 682 691 692 701 702 711 712 721 722 731 732 741 742 751 752 761 762 771 772 781 782 791 792 801 802 811 812 821 822 831 832 841 842 851 852 861 862 871 872 881 882 891 892 901 902 911 912 921 922 931 932 941 942 951 952 961 962 971 972 981 982 991 992

71. MEMBER RELEASES

72. 11 51 START MX MY MZ

73. 12 13 61 END MX MY MZ

75. LOAD 3 LIVE LOAD
76. *LL = 0.6 KPA
77. MEMBER LOAD
78. 3 TO 6 UNI GY -1.92
79. 11 11
80. MEMBER LOAD
81. 11 11
82. 1 61 UNI Y 4
83. 2 TO 4 UNI Y -4.0
84. 5 6 UNI Y -3.6
85. 5 6 UNI Y -3.6
86. LOAD COMB 1001 1.4DL + 1.6LL
87. 1 1.4 2 1.4 3 1.6
88. LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL
89. 1 1.2 2 1.2 3 1.2 11 1.2
90. 1 1.2 2 1.2 3 1.2 11 1.2
91. LOAD COMB 2001 1.0DL + 1.0LL
92. 1 1.0 2 1.0 3 1.0
93. LOAD COMB 2011 DL + LL + WL
94. 1 1.0 2 1.0 3 1.0 11 1.0
95. PERFORM ANALYSIS
96. STAAD PLANE

--- PAGE NO. 3

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 13/ 16/ 4
ORIGINAL/FINAL BAND-WIDTH= 9/ 3/
TOTAL DEGREES OF FREEDOM = 29
TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 29
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQD/AVAIL. DISK SPACE = 12.0/ 359483.5 MB

**START ITERATION NO. 2
**START ITERATION NO. 3
**NOTE-Tension/Compression converged after 3 iterations, Case= 1

**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 2

**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 3

**START ITERATION NO. 2
**NOTE-Tension/Compression converged after 2 iterations, Case= 11

97. LOAD LIST ALL
98. PRINT SUPPORT REACTION
99. STAAD PLANE

--- PAGE NO. 4

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = PLANE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
61	1	0.47	5.05	0.00	0.00	0.00	0.00
	2	1.91	16.16	0.00	0.00	0.00	0.00
	3	0.23	1.16	0.00	0.00	0.00	0.00
	11	9.94	2.47	0.00	0.00	0.00	0.00
	1001	3.70	31.54	0.00	0.00	0.00	0.00
	1011	15.06	29.81	0.00	0.00	0.00	0.00
	2001	2.61	22.36	0.00	0.00	0.00	0.00
	2011	12.55	24.84	0.00	0.00	0.00	0.00
62	1	-0.18	2.10	0.00	0.00	0.00	0.00
	2	0.00	-0.07	0.00	0.00	0.00	0.00
	3	0.00	-0.07	0.00	0.00	0.00	0.00
	11	6.82	-11.56	0.00	0.00	0.00	0.00
	1001	-1.56	-9.60	0.00	0.00	0.00	0.00
	1011	6.85	-9.60	0.00	0.00	0.00	0.00

Page 2

71. MEMBER RELEASES
72. 11 51 START MX MY MZ
73. 12 13 61 END MX MY MZ

Page 1

BRU Steel_Roof-TPS-ANL									
MEMB.	FY	DIST	LD	FZ	DIST	LD	FX	DIST	LD
MIN.	0.00	0.00	1	0.00	0.00	1	5.49 C	0.00	1001
	-0.13	1.04	3	-1.41	1.04	1001			
	0.00	1.04	2011	0.00	1.04	2011			

3. AMONGST ALL SECT LOCATIONS									
MEMB.	FY	DIST	LD	FZ	DIST	LD	FX	DIST	LD
MAX.	5.30	0.00	1011	0.71	0.00	11	1.28 C	0.00	3
MIN.	-2.87	1.04	1001	-2.18	0.62	1011	12.34 T	1.04	1011

4	0.00	MAX	-0.55	1	0.20	3	0.00	2011	0.00	2011
		MIN	-7.53	1011	-1.70	1011	0.00	2011	0.00	2011
	0.21	MAX	-0.65	1	0.81	1001	0.00	2011	0.00	2011
		MIN	-9.54	1011	-0.46	11	0.00	2011	0.00	2011
	0.42	MAX	-0.75	1	2.26	1001	0.00	2011	0.00	2011
		MIN	-11.55	1011	0.18	1	0.00	2011	0.00	2011
	0.62	MAX	-0.85	1	4.37	1011	0.00	2011	0.00	2011
		MIN	-13.55	1011	0.34	1	0.00	2011	0.00	2011
	0.83	MAX	-0.95	1	7.89	1011	0.00	2011	0.00	2011
		MIN	-11.05	1	11.33	1011	0.00	2011	0.00	2011
	1.04	MAX	-1.05	1	0.74	1	0.00	2011	0.00	2011
		MIN	-17.57	1011			0.00	2011	0.00	2011
STAAD PLANE							--PAGE NO.	7		

5. AMONGST ALL SECT LOCATIONS									
MEMB.	FY	DIST	LD	FZ	DIST	LD	FX	DIST	LD
MAX.	-0.55	0.00	1	11.33	1.04	1011	0.20 T	0.00	3
MIN.	-17.57	1.04	1011	-1.70	0.00	1011	13.97 T	1.04	1011

5	0.00	MAX	20.45	1011	11.33	1011	0.00	2011	0.00	2011
		MIN	1.21	1	0.74	1	0.00	2011	0.00	2011
	0.21	MAX	18.32	1011	7.31	1011	0.00	2011	0.00	2011
		MIN	1.08	1	0.50	1	0.00	2011	0.00	2011
	0.42	MAX	16.19	1011	3.73	1011	0.00	2011	0.00	2011
		MIN	0.36	1011	0.28	1011	0.00	2011	0.00	2011
	0.62	MAX	10.84	1011	-0.17	1011	0.00	2011	0.00	2011
		MIN	0.89	1	-0.13	3	0.00	2011	0.00	2011
	0.83	MAX	11.93	1011	-0.06	1	0.00	2011	0.00	2011
		MIN	0.72	1	-2.11	1011	0.00	2011	0.00	2011
	1.04	MAX	9.80	1011	-0.20	1	0.00	2011	0.00	2011
		MIN	0.60	1	-4.36	1011	0.00	2011	0.00	2011

7. AMONGST ALL SECT LOCATIONS									
MEMB.	FY	DIST	LD	FZ	DIST	LD	FX	DIST	LD
MAX.	20.45	0.00	1011	11.33	0.00	1011	4.90 C	0.00	1011
MIN.	0.60	1.04	1	-4.36	1.04	1011	0.14 C	1.04	1

BRU Steel_Roof-TPS-ANL									
MEMB.	FY	DIST	LD	FZ	DIST	LD	FX	DIST	LD
MIN.	0.00	0.00	1	0.00	0.00	1	15.79 C	0.00	1001
	-0.16	0.00	0.00	-10.28	2.97	1011	6.19 T	2.97	11
	0.00	2.97	2011	0.00	2.97	2011			

1. AMONGST ALL SECT LOCATIONS									
MEMB.	FY	DIST	LD	FZ	DIST	LD	FX	DIST	LD
MAX.	9.42	2.97	1011	1.02	0.59	11	15.79 C	0.00	1001
MIN.	-2.91	0.00	1	-10.28	2.97	1011	6.19 T	2.97	11

2. AMONGST ALL SECT LOCATIONS									
MEMB.	FY	DIST	LD	FZ	DIST	LD	FX	DIST	LD
MAX.	12.94	1011	3	10.28	1011	0.00	2011	0.00	2011
MIN.	-0.13	1011	3	-0.60	1001	0.00	2011	0.00	2011

3. AMONGST ALL SECT LOCATIONS										
MEMB.	FY	DIST	LD	FZ	DIST	LD	FX	DIST	LD	
MAX.	0.21	MAX	11.87	1011	7.71	1011	0.00	2011	0.00	2011
MIN.	-0.13	1011	3	-0.79	1001	0.00	2011	0.00	2011	
MEMB.	FY	DIST <td>LD<td>FZ</td><td>DIST<td>LD<td>FX</td><td>DIST<td>LD</td></td></td></td></td>	LD <td>FZ</td> <td>DIST<td>LD<td>FX</td><td>DIST<td>LD</td></td></td></td>	FZ	DIST <td>LD<td>FX</td><td>DIST<td>LD</td></td></td>	LD <td>FX</td> <td>DIST<td>LD</td></td>	FX	DIST <td>LD</td>	LD	
1	1.16	1001	0.00	2011	0.00	2011	0.00	2011	0.00	2011
	-2.91	11	0.00	2011	0.00	2011	0.00	2011	0.00	2011
0.59	0.62	1001	1.02	11	0.00	2011	0.00	2011	0.00	2011
	-0.53	11	-0.53	1001	0.00	2011	0.00	2011	0.00	2011
1.19	2.27	1011	0.63	11	0.00	2011	0.00	2011	0.00	2011
	0.00	1011	-0.73	1001	0.00	2011	0.00	2011	0.00	2011
1.78	0.00	1011	0.00	1011	0.00	2011	0.00	2011	0.00	2011
	-0.47	1001	-0.90	1011	0.00	2011	0.00	2011	0.00	2011
2.38	7.04	1011	0.00	3	0.00	2011	0.00	2011	0.00	2011
	-1.02	1001	-5.40	1011	0.00	2011	0.00	2011	0.00	2011
2.97	9.42	1011	0.60	1001	0.00	2011	0.00	2011	0.00	2011
	-1.57	1001	-10.28	1011	0.00	2011	0.00	2011	0.00	2011

4. AMONGST ALL SECT LOCATIONS									
MAX/MIN FORCE VALUES FOR MEMB					1. AMONGST ALL SECT LOCATIONS				
	FY	DIST	LD	FZ	DIST	LD	FX	DIST	LD
MAX.	0.42	2.97	1011	1.02	0.59	11			
	0.00	0.00	11	0.00	0.00	11			
MIN.	-2.91	0.00	11	-10.28	2.97	1011	15.79	C	0.00
	-0.00	2.97	2011	0.00	2.97	2011	6.19	T	2.97
									11

2	0.00	MAX	12.94	1011	10.28	1011	0.00	2011	0.00	2011
	0.21	MIN	-1.65	1013	-9.90	1001	0.00	2011	0.00	2011
		MAX	10.81	1011	-0.73	1001	0.00	2011	0.00	2011
	0.42	MIN	-0.13	1011	5.36	1001	0.00	2011	0.00	2011
		MAX	-0.13	1011	-0.97	1001	0.00	2011	0.00	2011
	0.62	MIN	9.74	1011	3.51	11	0.00	2011	0.00	2011
		MAX	-0.13	3	-1.13	1001	0.00	2011	0.00	2011
	0.83	MIN	8.68	1011	2.02	11	0.00	2011	0.00	2011
		MAX	-0.13	3	-1.28	1001	0.00	2011	0.00	2011
	1.04	MIN	7.61	1011	-0.71	11	0.00	2011	0.00	2011
		MAX	-0.13	3	-1.41	1001	0.00	2011	0.00	2011
		STAND PLANE						PAGE NO.		

6. AMONGST ALL SECT LOCATIONS									
MEMB.	FY	DIST	LD	FZ	DIST	LD	FX	DIST	LD
MAX.	12.94	0.00	1011	10.28	0.00	1011			
MIN.	-0.13	1011	3	-0.60	1001	0.00	2011	0.00	2011

MAX/MIN FORCE VALUES FOR MEMB		13. AMONGST ALL SECT LOCATIONS	
FY/	DIST LD	MZ/	DIST LD
1.14	MAX	0.01	2
	MIN	-0.02	11
1.71	MAX	0.01	2
	MIN	-0.02	11
2.27	MAX	0.01	2
	MIN	-0.02	11
2.84	MAX	0.01	2
	MIN	-0.02	11

MAX/MIN FORCE VALUES FOR MEMB		MZ/ DIST LD		13, AMONGST ALL SECT LOCATIONS	
FY/	DIST LD	FY/	DIST LD	FY/	DIST LD
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9
10	10	10	10	10	10
11	11	11	11	11	11
12	12	12	12	12	12
13	13	13	13	13	13
14	14	14	14	14	14
15	15	15	15	15	15
16	16	16	16	16	16
17	17	17	17	17	17
18	18	18	18	18	18
19	19	19	19	19	19
20	20	20	20	20	20
21	21	21	21	21	21
22	22	22	22	22	22
23	23	23	23	23	23
24	24	24	24	24	24
25	25	25	25	25	25
26	26	26	26	26	26
27	27	27	27	27	27
28	28	28	28	28	28
29	29	29	29	29	29
30	30	30	30	30	30
31	31	31	31	31	31
32	32	32	32	32	32
33	33	33	33	33	33
34	34	34	34	34	34
35	35	35	35	35	35
36	36	36	36	36	36
37	37	37	37	37	37
38	38	38	38	38	38
39	39	39	39	39	39
40	40	40	40	40	40
41	41	41	41	41	41
42	42	42	42	42	42
43	43	43	43	43	43
44	44	44	44	44	44
45	45	45	45	45	45
46	46	46	46	46	46
47	47	47	47	47	47
48	48	48	48	48	48
49	49	49	49	49	49
50	50	50	50	50	50
51	51	51	51	51	51
52	52	52	52	52	52
53	53	53	53	53	53
54	54	54	54	54	54
55	55	55	55	55	55
56	56	56	56	56	56
57	57	57	57	57	57
58	58	58	58	58	58
59	59	59	59	59	59
60	60	60	60	60	60
61	61	61	61	61	61
62	62	62	62	62	62
63	63	63	63	63	63
64	64	64	64	64	64
65	65	65	65	65	65
66	66	66	66	66	66
67	67	67	67	67	67
68	68	68	68	68	68
69	69	69	69	69	69
70	70	70	70	70	70
71	71	71	71	71	71
72	72	72	72	72	72
73	73	73	73	73	73
74	74	74	74	74	74
75	75	75	75	75	75
76	76	76	76	76	76
77	77	77	77	77	77
78	78	78	78	78	78
79	79	79	79	79	79
80	80	80	80	80	80
81	81	81	81	81	81
82	82	82	82	82	82
83	83	83	83	83	83
84	84	84	84	84	84
85	85	85	85	85	85
86	86	86	86	86	86
87	87	87	87	87	87
88	88	88	88	88	88
89	89	89	89	89	89
90	90	90	90	90	90
91	91	91	91	91	91
92	92	92	92	92	92
93	93	93	93	93	93
94	94	94	94	94	94
95	95	95	95	95	95
96	96	96	96	96	96
97	97	97	97	97	97
98	98	98	98	98	98
99	99	99	99	99	99
100	100	100	100	100	100

[illegible]

STAAD PLANE		-- PAGE NO. 10	
MIN	0.00 2011	0.00 2011	0.00 2011
2.53 MAX	0.00 2011	0.00 2011	0.00 2011
MIN	0.00 2011	0.00 2011	0.00 2011
3.38 MIN	0.00 2011	0.00 2011	0.00 2011
MIN	0.00 2011	0.00 2011	0.00 2011
4.22 MAX	0.00 11	0.00 2011	0.00 2011
MIN	-0.19 1081	0.00 2011	0.00 2011

MAX/MIN FORCE VALUES FOR MEMB		21. AMONGST ALL SECT LOCATIONS	
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[illegible]

MAX/MIN FORCE VALUES FOR MEMB		22. AMONGST ALL SECT		LOCATIONS	
		FV/	LD	DIST	LD
1.76	MAX	0.00	2011	0.00	2011
	MIN	0.00	2011	0.00	2011
2.64	MAX	0.00	2011	0.00	2011
	MIN	0.00	2011	0.00	2011
3.51	MAX	0.00	2011	0.00	2011
	MIN	0.00	2011	0.00	2011
4.39	MAX	0.00	2011	0.00	2011
	MIN	0.00	2011	0.00	2011

[illegible]

MIN	-15.75	1001	-0.53	11	0.00	2011
MAX	6.66	11	1.89	1001	0.00	2011
MIN	-15.78	1001	-0.80	11	0.00	2011
MAX	6.66	11	2.52	1001	0.00	2011
MIN	-15.82	1001	-1.07	11	0.00	2011

MAX/MIN FORCE VALUES FOR MEMB										71. AMONGST ALL SECT LOCATIONS									
		FY/		FZ/		DIST		LD		M2/		DIST		LD		FX		DIST	
		MAX		MIN		MAX		MIN		MAX		MIN		MAX		MIN		MAX	
		0.20		MAX		6.66		11		3.15		1001		0.00		2011		0.00	
		MIN		MIN		-15.85		1001		-1.33		11		0.00		2011		0.00	

STAAD PLANE

MAX/MIN FORCE VALUES FOR MEMB										51. AMONGST ALL SECT LOCATIONS									
		FY/		FZ/		DIST		LD		M2/		DIST		LD		FX		DIST	
		MAX		MIN		MAX		MIN		MAX		MIN		MAX		MIN		MAX	
		0.66		0.00		0.00		11		3.15		0.20		1001		3.67		0.00	
		MIN		MIN		-15.85		1001		-1.33		11		0.00		2011		0.00	

MAX/MIN FORCE VALUES FOR MEMB										52. AMONGST ALL SECT LOCATIONS									
		FY/		FZ/		DIST		LD		M2/		DIST		LD		FX		DIST	
		MAX		MIN		MAX		MIN		MAX		MIN		MAX		MIN		MAX	
		0.00		MAX		19.01		1011		3.15		1001		0.00		2011		0.00	
		MIN		MIN		-1.33		11		-1.33		11		0.00		2011		0.00	

MAX/MIN FORCE VALUES FOR MEMB										52. AMONGST ALL SECT LOCATIONS									
		FY/		FZ/		DIST		LD		M2/		DIST		LD		FX		DIST	
		MAX		MIN		MAX		MIN		MAX		MIN		MAX		MIN		MAX	
		0.00		MAX		19.01		0.00		0.00		1		6.27		C		0.00	
		MIN		MIN		-3.86		0.61		1001		11		0.76		T		0.61	

MAX/MIN FORCE VALUES FOR MEMB										61. AMONGST ALL SECT LOCATIONS									
		FY/		FZ/		DIST		LD		M2/		DIST		LD		FX		DIST	
		MAX		MIN		MAX		MIN		MAX		MIN		MAX		MIN		MAX	
		0.00		MAX		1.48		1001		1.12		1001		0.00		2011		0.00	
		MIN		MIN		-10.02		1011		-6.93		1011		0.00		2011		0.00	

MAX/MIN FORCE VALUES FOR MEMB										61. AMONGST ALL SECT LOCATIONS									
		FY/		FZ/		DIST		LD		M2/		DIST		LD		FX		DIST	
		MAX		MIN		MAX		MIN		MAX		MIN		MAX		MIN		MAX	
		0.00		MAX		1.48		1001		1.12		1001		0.00		2011		0.00	
		MIN		MIN		-10.02		1011		-6.93		1011		0.00		2011		0.00	

MAX/MIN FORCE VALUES FOR MEMB										61. AMONGST ALL SECT LOCATIONS									
		FY/		FZ/		DIST		LD		M2/		DIST		LD		FX		DIST	
		MAX		MIN		MAX		MIN		MAX		MIN		MAX		MIN		MAX	
		0.00		MAX		1.48		1001		1.12		1001		0.00		2011		0.00	
		MIN		MIN		-10.02		1011		-6.93		1011		0.00		2011		0.00	

MAX/MIN FORCE VALUES FOR MEMB										71. AMONGST ALL SECT LOCATIONS									
		FY/		FZ/		DIST		LD		M2/		DIST		LD		FX		DIST	
		MAX		MIN		MAX		MIN		MAX		MIN		MAX		MIN		MAX	
		0.00		MAX		1.04		0.00		6.93		0.00		1011		3.93		C	
		MIN		MIN		-0.80		3.80		1001		1		7.39		T		3.80	

102. PARAMETER 1
103. CODE B55950
104. PY 345000 ALL
105. CHECK CODE ALL
106. CHECK CODE ALL
STEEL DESIGN

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.9.5950-1_2000
STAAD PLANE

--- PAGE NO. 15

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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1	TAPERED	PASS	BS-4.8.2.2	0.033	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2164.0 PT= 2693.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

2	TAPERED	PASS	BS-4.8.2.2	0.032	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2677.2 PT= 2693.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

3	TAPERED	PASS	BS-4.2.3-(Y)	0.011	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2677.1 PT= 2693.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

4	TAPERED	PASS	BS-4.8.2.2	0.039	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 11908.2 PT= 2695.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

5	TAPERED	PASS	BS-4.2.3-(Y)	0.041	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2677.3 PT= 2695.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

STAAD PLANE					
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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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1	TAPERED	PASS	BS-4.8.2.2	0.033	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2164.0 PT= 2693.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

2	TAPERED	PASS	BS-4.8.2.2	0.032	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2677.2 PT= 2693.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

3	TAPERED	PASS	BS-4.2.3-(Y)	0.011	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2677.1 PT= 2693.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

4	TAPERED	PASS	BS-4.8.2.2	0.039	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 11908.2 PT= 2695.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

5	TAPERED	PASS	BS-4.2.3-(Y)	0.041	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2677.3 PT= 2695.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

STAAD PLANE					
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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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1	TAPERED	PASS	BS-4.8.2.2	0.033	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2164.0 PT= 2693.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

2	TAPERED	PASS	BS-4.8.2.2	0.032	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2677.2 PT= 2693.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

3	TAPERED	PASS	BS-4.2.3-(Y)	0.011	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2677.1 PT= 2693.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

4	TAPERED	PASS	BS-4.8.2.2	0.039	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 11908.2 PT= 2695.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

5	TAPERED	PASS	BS-4.2.3-(Y)	0.041	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2677.3 PT= 2695.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

STAAD PLANE					
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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
--------	-------	---------------	----------------------	--------------	----------------------

1	TAPERED	PASS	BS-4.8.2.2	0.033	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2164.0 PT= 2693.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

2	TAPERED	PASS	BS-4.8.2.2	0.032	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2677.2 PT= 2693.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

3	TAPERED	PASS	BS-4.2.3-(Y)	0.011	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2677.1 PT= 2693.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

4	TAPERED	PASS	BS-4.8.2.2	0.039	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 11908.2 PT= 2695.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

5	TAPERED	PASS	BS-4.2.3-(Y)	0.041	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2677.3 PT= 2695.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

STAAD PLANE					
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BRU Steel_Roof-TPS.ANL
: BS-4.4.3.3 status = PASS

6	TAPERED	PASS	BS-4.7	0.030	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2346.9 PT= 2693.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

11	TAPERED	PASS	BS-4.7	0.038	2011
CALCULATED CAPACITIES FOR MEMB MCZ= 270.6 MCY= 111.6 PC= 2583.3 PT= 2693.8 MB= 270.6 BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00 Shear Buckling check is required : Vb = 332 kN : a = 0 mm : pyf = 207 N/mm ² d = 200 mm : t = 8 mm : a = 0 mm : pyf = 344 N/mm ² BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

12	ST	TUB1501506.3	PASS	BS-4.8.3.3	0.034	2011
MATERIAL DATA Grade of steel = S 275 Modulus of elasticity = 205 kN/mm ² Design strength (py) = 344 N/mm ² SECTION PROPERTIES (units - cm) Member Length = 256.70 Gross Area = 35.80 Net Area = 35.80 Eff. Area = 35.80 Moment of inertia : Z-Z axis : 1223.000 Y-Y axis : 1223.000 Plastic modulus : 192.000 Elastic modulus : 163.067 Effective modulus : 163.067 Shear Area : 17.900 DESIGN DATA (units - KN.m) Section Class : SEMI-COMPACT Squash Load : 1235.10 Axial force/squash load : 0.030 Compression Capacity : Z-Z axis : 1141.5 Y-Y axis : 1141.5 Moment Capacity : 66.2 Reduced Moment Capacity : 66.2 Shear Capacity : 370.5 BUCKLING CALCULATIONS (units - KN.m) (axis nomenclature as per design code) Slenderness : X-X axis : 43.919 Y-Y axis : 43.919 Radius of gyration (cm) : 5.845 Effective Length : 2.567						

LTB check unnecessary for this section

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- KN.m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MY	MZ	NY
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BS-4.8.3.2	0.033	2011	37.6	0.0	0.0	0.0	0.0	0.0
BS-4.8.3.3	0.034	2011	37.6	0.0	0.0	0.0	0.0	0.0
BS-4.8.3.3	0.034	2011	37.6	0.0	0.0	0.0	0.0	0.0

Torsion and deflections have not been considered in the design.

STAAD PLANE

--- PAGE NO. 17

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
--------	-------	---------------	----------------------	--------------	----------------------

12	ST	TUB1501506.3	PASS	BS-4.8.3.3	0.034	2011
MATERIAL DATA Grade of steel = S 275 Modulus of elasticity = 205 kN/mm ² Design strength (py) = 344 N/mm ² SECTION PROPERTIES (units - cm) Member Length = 256.70 Gross Area = 35.80 Net Area = 35.80 Eff. Area = 35.80 Moment of inertia : Z-Z axis : 1223.000 Y-Y axis : 1223.000 Plastic modulus : 192.000 Elastic modulus : 163.067 Effective modulus : 163.067 Shear Area : 17.900 DESIGN DATA (units - KN.m) Section Class : SEMI-COMPACT Squash Load : 1235.10 Axial force/squash load : 0.030 Compression Capacity : Z-Z axis : 1141.5 Y-Y axis : 1141.5 Moment Capacity : 66.2 Reduced Moment Capacity : 66.2 Shear Capacity : 370.5 BUCKLING CALCULATIONS (units - KN.m) (axis nomenclature as per design code) Slenderness : X-X axis : 43.919 Y-Y axis : 43.919 Radius of gyration (cm) : 5.845 Effective Length : 2.567						

LTB check unnecessary for this section

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- KN.m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MY	MZ	NY
--------	-------	------	----	----	----	----	----	----

BS-4.8.3.2	0.033	2011	37.6	0.0	0.0	0.0	0.0	0.0
BS-4.8.3.3	0.034	2011	37.6	0.0	0.0	0.0	0.0	0.0
BS-4.8.3.3	0.034	2011	37.6	0.0	0.0	0.0	0.0	0.0

Torsion and deflections have not been considered in the design.

STAAD PLANE

--- PAGE NO. 18

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

DESIGN DATA (units - kn,m)
Section Class : BS5950-1/2000
SLENDER

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
--------	-------	---------------	----------------------	--------------	----------------------

13 ST TUB1501506.3	PASS	13.59 C	BS-4.8.3.3.3	0.013	2011
			0.00	0.03	-

MATERIAL DATA (see)

Grade of steel	= S 275
Modulus of elasticity	= 205 kN/mm ²
Design Strength (py)	= 344 N/mm ²

SECTION PROPERTIES (units - cm)

Member Length	= 284.20
Gross Area	= 35.80
Net Area	= 35.80
Eff. Area	= 35.80

Moment of inertia	: 1223.000	z-z axis	y-y axis
Plastic modulus	: 192.000		
Elastic modulus	: 163.067		
Effective modulus	: 163.067		
Shear Area	: 17.900		

DESIGN DATA (units - kn,m)

Section Class	: BS5950-1/2000
Slenderness	: 123.00
Axial force/squash load	: 0.011

Compression Capacity	: 1117.9	z-z axis	y-y axis
Moment Capacity	: 66.2		
Reduced Moment Capacity	: 66.2		
Shear Capacity	: 370.5		

BUCKLING CALCULATIONS (units - kn,m)

(axis nomenclature as per design code)

Slenderness	: 48.624	x-x axis	y-y axis
Radius of gyration (cm)	: 5.845		
Effective Length	: 2.842		

LTB check unnecessary for this section

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kn,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY	LOADING/ LOCATION
BS-4.7 (C)	0.012	2011	13.6	0.0	0.0	0.0	0.0	
BS-4.8.3.2	0.011	2011	13.1	0.0	0.0	0.0	0.0	
BS-4.8.3.3.1	0.013	2011	13.6	-	-	0.0	0.0	
BS-4.8.3.3.3	0.013	2011	13.6	-	-	0.0	0.0	

Torsion and deflections have not been considered in the design.

STAAD PLANE

STAAD PLANE

-- PAGE NO. 19

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

DESIGN DATA (units - kn,m)
Section Class : BS5950-1/2000
SLENDER

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
--------	-------	---------------	----------------------	--------------	----------------------

21 ST UA90X90X6	PASS	0.87 T	BS-4.6	0.002	2001
			0.00	0.00	4.22

MATERIAL DATA (see)

Grade of steel	= S 275
Modulus of elasticity	= 205 kN/mm ²
Design Strength (py)	= 344 N/mm ²

SECTION PROPERTIES (units - cm)

Member Length	= 421.96
Gross Area	= 10.60
Net Area	= 10.60
Eff. Area	= 10.60

Moment of inertia	: 131.618	z-z axis	y-y axis
Plastic modulus	: 37.227		
Elastic modulus	: 20.682		
Effective modulus	: 20.682		
Shear Area	: 3.240		

Page 11

DESIGN DATA (units - kn,m)
Section Class : BS5950-1/2000
SLENDER

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
--------	-------	---------------	----------------------	--------------	----------------------

22 ST UA90X90X6	PASS	1.47 T	BS-4.6	0.004	2011
			0.00	0.00	0.00

MATERIAL DATA (see)

Grade of steel	= S 275
Modulus of elasticity	= 205 kN/mm ²
Design Strength (py)	= 344 N/mm ²

SECTION PROPERTIES (units - cm)

Member Length	= 439.24
Gross Area	= 10.60
Net Area	= 10.60
Eff. Area	= 10.60

Moment of inertia	: 131.618	z-z axis	y-y axis
Plastic modulus	: 37.227		
Elastic modulus	: 20.682		
Effective modulus	: 20.682		
Shear Area	: 3.240		

DESIGN DATA (units - kn,m)

Section Class	: BS5950-1/2000
Slenderness	: 123.00
Axial force/squash load	: 0.011

Compression Capacity	: 1117.9	z-z axis	y-y axis
Moment Capacity	: 66.2		
Reduced Moment Capacity	: 66.2		
Shear Capacity	: 370.5		

BUCKLING CALCULATIONS (units - kn,m)

(axis nomenclature as per design code)

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY	LOADING/ LOCATION
BS-4.7 (C)	0.012	2011	13.6	0.0	0.0	0.0	0.0	
BS-4.8.3.2	0.011	2011	13.1	0.0	0.0	0.0	0.0	
BS-4.8.3.3.1	0.013	2011	13.6	-	-	0.0	0.0	
BS-4.8.3.3.3	0.013	2011	13.6	-	-	0.0	0.0	

Torsion and deflections have not been considered in the design.

STAAD PLANE

STAAD PLANE

-- PAGE NO. 20

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

DESIGN DATA (units - kn,m)
Section Class : BS5950-1/2000
SLENDER

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
--------	-------	---------------	----------------------	--------------	----------------------

21 ST UA90X90X6	PASS	0.87 T	BS-4.6	0.002	2001
			0.00	0.00	4.22

MATERIAL DATA (see)

Grade of steel	= S 275
Modulus of elasticity	= 205 kN/mm ²
Design Strength (py)	= 344 N/mm ²

SECTION PROPERTIES (units - cm)

Member Length	= 421.96
Gross Area	= 10.60
Net Area	= 10.60
Eff. Area	= 10.60

Moment of inertia	: 131.618	z-z axis	y-y axis
Plastic modulus	: 37.227		
Elastic modulus	: 20.682		
Effective modulus	: 20.682		
Shear Area	: 3.240		

Page 12

CALCULATED CAPACITIES FOR MEMB 51 UNIT - kN,m SECTION CLASS 3									
MCZ=	270.6	MCY=	111.6	PC=	2693.8	PT=	0.0	MB=	270.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mX = 1.00, mY = 1.00, mZ = 1.00									
Shear Buckling check is required : Vb = 332 kN : qw = 207 N/mm2									
d =	200 mm	t =	8 mm	a =	0 mm	pyf =	344 N/mm2		
BS-4.4.3.2 status = PASS									
BS-4.4.3.2 : BS-4.4.3.3 status = PASS									
52	TAPERED	PASS	BS-4.2.3-(Y)	0.038	2011				
STAAD PLANE									
CALCULATED CAPACITIES FOR MEMB 52 UNIT - kN,m SECTION CLASS 3									
MCZ=	270.6	MCY=	111.6	PC=	2693.8	PT=	2693.8	MB=	270.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mX = 1.00, mY = 1.00, mZ = 1.00									
Shear Buckling check is required : Vb = 332 kN : qw = 207 N/mm2									
d =	200 mm	t =	8 mm	a =	0 mm	pyf =	344 N/mm2		
BS-4.4.3.2 status = PASS									
BS-4.4.3.2 : BS-4.4.3.3 status = PASS									
STAAD PLANE									
PAGE NO.									

--- PAGE NO. 22

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/	CRITICAL COND/	RATIO/	LOADING/
		FX	MY	MZ	LOCATION

61 ST TUB1401406.0 PASS ANNEX I-1 0.111 0.111 2011

--- PAGE NO. 22

MATERIAL DATA

Grade of steel = S 275

Modulus of elasticity = 205 kN/mm2

Design strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 384.80

Gross Area = 31.80 Net Area = 31.80 Eff. Area = 31.80

Moment of inertia : Z-Z axis Y-Y axis

Plastic modulus : 944.000 844.000

Elastic modulus : 159.000 159.000

Effective modulus : 134.857 134.857

Shear Area : 15.900 15.900

DESIGN DATA (units - KN.m)

Section Class : BS5950-1/2000

Squash Load : PLASTIC

Axial force/squash load : 1097.10

Compression Capacity : Z-Z axis Y-Y axis

Moment Capacity : 835.1 835.1

Reduced Moment Capacity : 54.9 54.9

Shear Capacity : 329.1 329.1

BUCKLING CALCULATIONS (units - KN.m)

(axis nomenclature as per design code)

Slenderness : X-X axis Y-Y axis

Radius of gyration (cm) : 70.626 70.626

Effective Length : 5.448 5.448

LTB check unnecessary for this section

CRITICAL LOADS FOR EACH CLAUSE CHECK (units - KN.m) :

BS-4.2.3-(Y) RATIO LOAD FX VY VZ MZ MY

BS-4.2.3-(Y) 0.025 2011 30.4 8.3 - -

BS-4.8.3.2 0.026 2011 30.4 8.3 0.0 5.8 0.0

BS-4.8.3.3 0.161 2011 30.4 - - 5.8 0.0

BS-4.8.3.3.1 0.161 2011 30.4 - - 5.8 0.0

BS-4.8.3.3.3 0.144 2011 30.4 - - 5.8 0.0

ANNEX I-1 0.111 2011 30.4 - - 5.8 0.0

STAAD PLANE

--- PAGE NO. 23

Torsion and deflections have not been considered in the design.

71 TAPERED PASS BS-4.8.2.2 0.023 2011

3.88 T 0.00 5.78 0.00

Page 13

CALCULATED CAPACITIES FOR MEMB 72 UNIT - KN.M SECTION CLASS 3														
MCZ=	270.6	MCY=	111.6	PC=	1928.4	PT=	2693.8	MB=	270.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mX = 1.00, mY = 1.00, mZ = 1.00														
Shear Buckling check is required : Vb = 332 kN : qw = 207 N/mm2														
d =	200 mm	t =	8 mm	a =	0 mm	pyf =	344 N/mm2							
BS-4.4.3.2 status = PASS														
BS-4.4.3.2 : BS-4.4.3.3 status = PASS														
72	TAPERED	PASS	BS-4.7	0.00	0.050	2011								
STAAD PLANE														
CALCULATED CAPACITIES FOR MEMB 72 UNIT - KN.M SECTION CLASS 3														
MCZ=	270.6	MCY=	111.6	PC=	2384.5	PT=	2693.8	MB=	270.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mX = 1.00, mY = 1.00, mZ = 1.00														
Shear Buckling check is required : Vb = 332 kN : qw = 207 N/mm2														
d =	200 mm	t =	8 mm	a =	0 mm	pyf =	344 N/mm2							
BS-4.4.3.2 status = PASS														
BS-4.4.3.2 : BS-4.4.3.3 status = PASS														
72	TAPERED	PASS	BS-4.7	0.00	0.050	2011								
STAAD PLANE														

***** END OF TABULATED RESULT OF DESIGN *****

107. FINISH

***** END OF THE STAAD.Pro RUN *****

**** DATE= JUL 18,2007 TIME= 15:59:44 ****

3 ROOF BEAMS AND SIDE BEAM DESIGN

Document No.: **DM001-E-AAM-CDP-DR-DCC-373909**

Rev A4

Date: 21 Jul 2007

Page

Calculation/Sketc

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai Depot - Jebel Ali	Date	Jul '07	Job No.	068010
	TPS (Building 9)				
	Roof Secondary Beam Design	Designed by	Rudy	Sheet No.	

DL

Selfweight	0.40 x	1.00 =	0.4
Sheeting	0.10 x	3.80 =	0.38
Purlins	0.10 x	3.80 =	0.38
Insulation	0.30 x	3.80 =	1.14
M&E Finishes	0.30 x	3.80 =	1.14
			<u>3.44 kN/m</u>

LL

$$0.60 \times 3.80 = 2.28 \text{ kN/m}$$

$$w(sls) = 5.72 \text{ kN/m}$$

$$w(uls) = 8.5 \text{ kN/m}$$

a) For simply supported case with L = 3.6 m

$$M_{max} = 13.7 \text{ kNm}$$

$$V_{max} = 15.2 \text{ kN}$$

b) For cantilever case with L = 3.2 m

$$M_{max} = 43.3 \text{ kNm}$$

$$V_{max} = 27.1 \text{ kN}$$

Site	Dubai Depot - Jebel Ali	Date	Jul '07	Job No.	068010
	TPS (Building 9)				
	Roof Secondary Beam Design	Design	Rudy	Sheet No.	

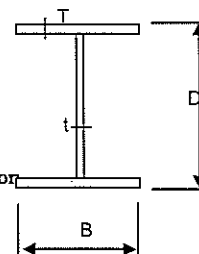
Steel Beam

- Check for M_b
- UB with equal flange
- For class 1 plastic, class 2 compact or class 3 semi-compact cross-sections

Section Properties

Build-up Section

Section	=	w	("r" for Rolled section, "w" for Welded section)
P_y	=	355	N/mm ²
E	=	205000	N/mm ²
D	=	250	mm
B	=	150	mm
t	=	6	mm
T	=	8	mm
$b/T = 9.0 < 13\epsilon$ $d/t = 39.0 < 80\epsilon$ ⇒ Class 3 semi-compact section			
B1 (Beam / Column Mark)			
A	=	38.04	cm ²
I_x	=	4155.77	cm ⁴
I_y	=	450.45	cm ⁴
Z_x	=	332.46	cm ³
Z_y	=	60.06	cm ³
S_x	=	372.53	cm ³
S_y	=	15.02	cm ³
r_x	=	10.45	cm
r_y	=	3.44	cm



Design Data

M_x	=	43.3	kNm
V	=	27.1	kN
L	=	3.2	m
L_e	=	4.32	m
			Overall span
			Effective length for M_b (1.6*2.7)

Design for Bending Moment

λ	=	126
From BS5950 clause. 4.3.6.7		
u	=	1.00
x	=	31.25
λ/x	=	4.02
v	=	0.86
Sect'n class	=	3.00
β_w	=	0.9
Equal flange { 1 for class 1 plastic 2 for class 2 compact 3 for class 3 semi-compact 4 for slender		
λ_{LT}	=	$uv\lambda(\beta_w)^{0.5} = 102$

From Table 16 (rolled section) or Table 17 (welded section)

p_b	=	97	N/mm ²
M_b	=	32.2	kNm

From Table 18

m_{LT}	=	0.6							
M_b/m_{LT}	=	53.7	kNm	>	M_x	=	43.3	kNm	OK !

Design for Bending Shear

A_v	=	1404	mm ²
P_v	=	299	kN
$0.6P_v$	=	179	kN
	>	V	= 27.1 kN
			Low shear, OK !

Deflection Check

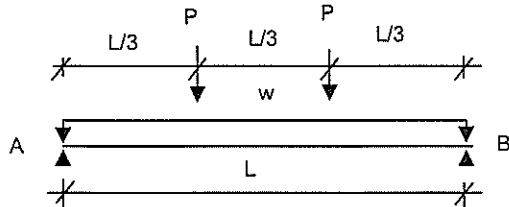
LL	=	2.3	kN/m	Uniform load
M_{LL}	=	11.8	kNm	
d	=	1.4	mm	< 25mm
d/L	=	1/2261		< 1/250
				OK!
				OK!

Calculation/Sketc

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai Depot - Jebel Ali TPS (Building 9) Roof Beam Design (Above BRU Enclosure)	Date	Jul '07	Job No.	068010
		Designed by	Rudy	Sheet No.	



From STAAD Pro Analysis

P = 42.2 kN (ULS)
38.0 kN (SLS)

w-selfwt = 0.8 kN/m (ULS)
0.6 kN/m (SLS)

L = 9.6 m

Reaction $R_a = R_b =$ 46.2 kN (ULS)
40.9 kN

At $x = L/3$, $M =$ 143.6 kNm (ULS)

Using 300 x 250 UB with $T =$ 12 mm
 $t =$ 6 mm

$E =$ 205000 N/mm²
 $I_x =$ 13500.03 cm⁴ = 135000288 mm⁴

Deflection due to point load
 $d_1 =$ 9.4 mm

Max deflection due to selfwt =
 $d_2 =$ 2.4 mm

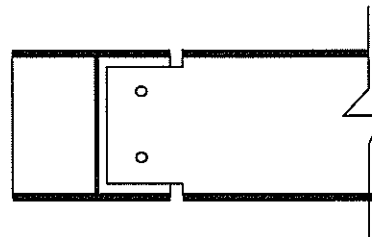
Therefore max deflection = 11.8 mm

Ok, less than 25 or $L/200$
($L/200 =$ 48 mm)

Bolt Connection Design

Shear, $F_y =$ 46.2 kN
Moment, $M_z =$ 0 kNm
Axial Force, $F_x =$ 0 kN
No of bolt provided = 2
Shear per bolt = 23 kN
Bolt diameter = 16 mm
Bolt's shear capacity = 58.9 kN

Ok





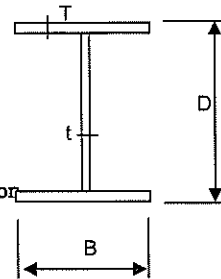
Site	Dubai Depot - Jebel Ali TPS (Building 9)	Date	Jul '07	Job No.	068010
	Roof Beam Design (Above BRU Enclosure)	Design	Rudy	Sheet No.	

Steel Beam

- Check for Mb
- UB with equal flange
- For class 1 plastic, class 2 compact or class 3 semi-compact cross-sections

Section Properties Build-up Section

Section	=	w	("r" for Rolled section, "w" for Welded section)			
p _y	=	355	N/mm2			
E	=	205000	N/mm2			
D	=	300	mm	} b/T = 10.2 < 13ε d/t = 46.0 < 80ε ⇒ Class 3 semi-		
B	=	250	mm			
t	=	6	mm			
T	=	12	mm			
			B1 (Beam / Column Mark)			
A	=	76.56	cm2	=	7656	mm2
I _x	=	13500.03	cm4	=	135000288	mm4
I _y	=	3125.54	cm4	=	31255400	mm4
Z _x	=	900.00	cm3	=	900002	mm3
Z _y	=	250.04	cm3	=	250043	mm3
S _x	=	978.26	cm3	=	978264	mm3
S _y	=	62.51	cm3	=	62510.8	mm3
r _x	=	13.28	cm	=	132.7903101	mm
r _y	=	6.39	cm	=	63.89421933	mm



Design Data

M _x	=	142.2	kNm	
V	=	46.2	kN	
L	=	9.6	m	Overall span
L _e	=	3.2	m	Effective length for Mb

Design for Bending Moment

λ	=	50	
From BS5950 clause. 4.3.6.7			
u	=	1.00	
x	=	25.00	
λ/x	=	2.00	
v	=	0.96	Equal flange
Sect'n class	=	3.00	1 for class 1 plastic 2 for class 2 compact 3 for class 3 semi-compact 4 for slender
β_w	=	0.9	

$$\lambda_{LT} = uv\lambda(\beta_w)^{0.5} = 46$$

From Table 16 (rolled section) or Table 17 (welded section)

p_b	=	270	N/mm ²
M _b	=	243.0	kNm

From Table 18

m_{LT}	=	0.6	
M _b /m _{LT}	=	405.0	kNm
	>	M _x	= 142.2 kNm OK !

Design for Bending Shear

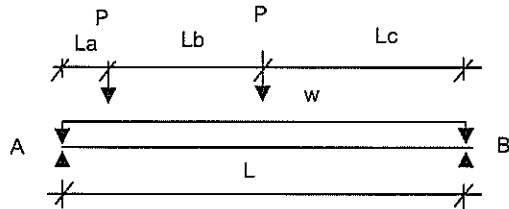
A _v	=	1656	mm ²	
P _v	=	353	kN	
0.6P _v	=	212	kN	> V = 46.2 kN Low shear, OK !

Calculation/Sketc

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai Depot - Jebel Ali TPS (Building 9) Roof Beam Design (Above BRU Enclosure)	Date	Jul '07	Job No.	068010
		Designed by	Rudy	Sheet No.	



From STAAD Pro Analysis

P = 46.2 kN (ULS)
38.0 kN (SLS)

w-selfwt = 0.8 kN/m (ULS)
0.6 kN/m (SLS)

L = 6.4 m
La = 0.6 m
Lb = 2.6 m
Lc = 3.2 m

Reaction Ra = 67.7 kN (ULS)
Rb = 30.1 kN (ULS)

at $x = L/2$, M = 174.6 kNm

Using 300 x 250 UB with T = 12 mm
t = 6 mm

E = 205000 N/mm²
Ix = 13500.03 cm⁴ = 135000288 mm⁴

Deflection due to point load
d1 = 9.6 mm

Max deflection due to selfwt =
d2 = 0.5 mm

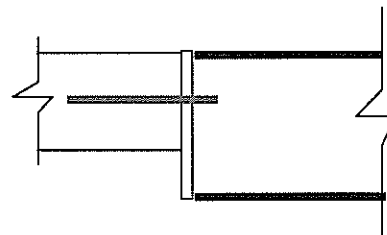
Therefore max deflection = 10.1 mm

Ok, less than 25 or L/200
(L/200 = 32 mm)

Bolt Connection Design

Shear, Fy = 67.7 kN
Moment, Mz = 0 kNm
Axial Force, Fx = 0 kN
No of bolt provided = 2
Shear per bolt = 34 kN
Bolt diameter = 16 mm
Bolt's shear capacity = 58.9 kN

Ok





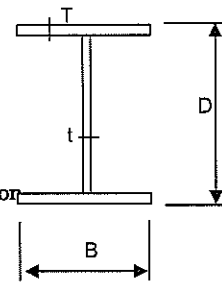
Site	Dubai Depot - Jebel Ali TPS (Building 9)	Date	Jul '07	Job No.	068010
	Roof Beam Design (Above BRU Enclosure)	Design	Rudy	Sheet No.	

Steel Beam

- Check for Mb
- UB with equal flange
- For class 1 plastic, class 2 compact or class 3 semi-compact cross-sections

Section Properties

		Build-up Section	
Section	=	w	("r" for Rolled section, "w" for Welded section)
p_y	=	355	N/mm ²
E	=	205000	N/mm ²
D	=	300	mm
B	=	250	mm
t	=	6	mm
T	=	12	mm
$b/T = 10.2 < 13\epsilon$ $d/t = 46.0 < 80\epsilon$ $\Rightarrow$ Class 3 semi-compact section			
B1 (Beam / Column Mark)			
A	=	76.56	cm ² = 7656 mm ²
I_x	=	13500.03	cm ⁴ = 135000288 mm ⁴
I_y	=	3125.54	cm ⁴ = 31255400 mm ⁴
Z_x	=	900.00	cm ³ = 900002 mm ³
Z_y	=	250.04	cm ³ = 250043 mm ³
S_x	=	978.26	cm ³ = 978264 mm ³
S_y	=	62.51	cm ³ = 62510.8 mm ³
r_x	=	13.28	cm = 132.7903101 mm
r_y	=	6.39	cm = 63.89421933 mm



Design Data

M_x	=	174.6	kNm
V	=	67.7	kN
L	=	6.4	m Overall span
L_e	=	3.2	m Effective length for Mb

Design for Bending Moment

λ	=	50	
From BS5950 clause. 4.3.6.7			
u	=	1.00	
x	=	25.00	
λ/x	=	2.00	
v	=	0.96	
Sect'n class	=	3.00	
β_w	=	0.9	
Equal flange { 1 for class 1 plastic 2 for class 2 compact 3 for class 3 semi-compact 4 for slender			
λ_{LT}	=	$uv\lambda(\beta_w)^{0.5}$	= 46
From Table 16 (rolled section) or Table 17 (welded section)			
p_b	=	270	N/mm ²
M_b	=	243.0	kNm
From Table 18			
m_{LT}	=	0.6	
M_b/m_{LT}	=	405.0	kNm
	>	M_x	= 174.6 kNm OK !

Design for Bending Shear

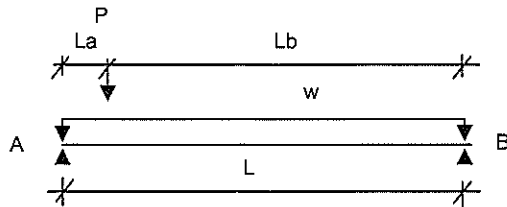
A_v	=	1656	mm ²
P_v	=	353	kN
$0.6P_v$	=	212	kN
	>	V	= 67.7 kN Low shear, OK !

Calculation/Sketc

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Site	Dubai Depot - Jebel Ali TPS (Building 9) Roof Beam Design (Above BRU Enclosure)	Date	Jul '07	Job No.	068010
		Designed by	Rudy	Sheet No.	



From STAAD Pro Analysis

P = 46.2 kN (ULS)
38.0 kN (SLS)

w-selfwt = 0.8 kN/m (ULS)
0.6 kN/m (SLS)

L = 3.2 m
La = 0.6 m
Lb = 2.6 m

Reaction Ra = 38.9 kN (ULS)
Rb = 8.5 kN (ULS)

at x = La, M = 23.2 kNm

Using 250 x 250 UB with T = 8 mm
t = 6 mm

E = 205000 N/mm²
Ix = 6499.18 cm⁴ = 64991785.3 mm⁴

Deflection due to point load
d1 = 3.0 mm

Max deflection due to selfwt =
d2 = 0.1 mm

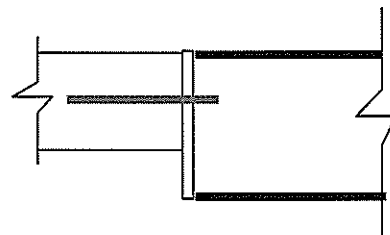
Therefore max deflection = 3.1 mm

Ok, less than 25 or L/200
(L/200 = 16 mm)

Bolt Connection Design

Shear, Fy = 38.9 kN
Moment, Mz = 0 kNm
Axial Force, Fx = 0 kN
No of bolt provided = 2
Shear per bolt = 19 kN
Bolt diameter = 16 mm
Bolt's shear capacity = 58.9 kN

Ok





Site	Dubai Depot - Jebel Ali TPS (Building 9)	Date	Jul '07	Job No.	068010
	Roof Beam Design (Above BRU Enclosure)	Design	Rudy	Sheet No.	

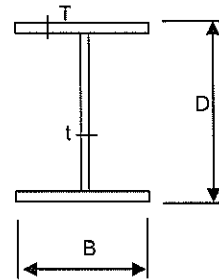
Steel Beam

- Check for M_b
- UB with equal flange
- For class 1 plastic, class 2 compact or class 3 semi-compact cross-sections

Section Properties

Build-up Section

Section	=	w	("r" for Rolled section, "w" for Welded section)	
p_y	=	355	N/mm2	
E	=	205000	N/mm2	
D	=	250	mm	} $b/T = 15.3 > 13\varepsilon$ $d/t = 39.0 < 80\varepsilon$ $\Rightarrow$ Slender section !
B	=	250	mm	
t	=	6	mm	
T	=	8	mm	
B1 (Beam / Column Mark)				
A	=	54.04	cm2	= 5404 mm2
I_x	=	6499.18	cm4	= 64991785.33 mm4
I_y	=	2083.78	cm4	= 20837833.33 mm4
Z_x	=	519.93	cm3	= 519934 mm3
Z_y	=	166.70	cm3	= 166703 mm3
S_x	=	566.13	cm3	= 566134 mm3
S_y	=	41.68	cm3	= 41675.66667 mm3
r_x	=	10.97	cm	= 109.665889 mm
r_y	=	6.21	cm	= 62.09671269 mm



Design Data

M_x	=	23.2	kNm	
V	=	38.9	kN	
L	=	3.2	m	Overall span
L_e	=	3.2	m	Effective length for M_b

Design for Bending Moment

λ	=	52	
From BS5950 clause. 4.3.6.7			
u	=	1.00	
x	=	31.25	
λ/x	=	1.65	
v	=	0.97	
Sect'n class	=	3.00	Equal flange { 1 for class 1 plastic 2 for class 2 compact 3 for class 3 semi-compact 4 for slender
β_w	=	0.9	

$$\lambda_{LT} = uv\lambda(\beta_w)^{0.5} = 48$$

From Table 16 (rolled section) or Table 17 (welded section)

p_b	=	253	N/mm ²
M_b	=	131.5	kNm

From Table 18

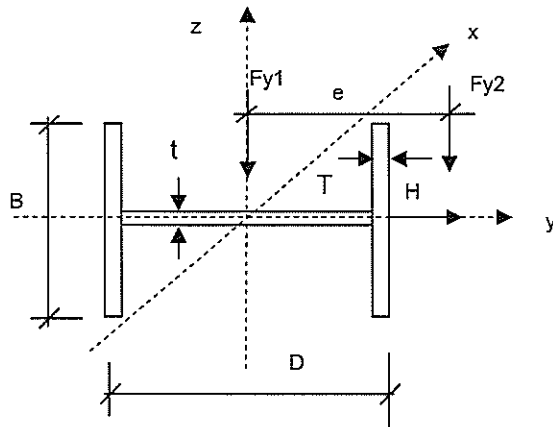
m_{LT}	=	0.6	
M_b/m_{LT}	=	219.2	kNm
	>	M_x	= 23.2 kNm OK !

Design for Bending Shear

A_v	=	1404	mm ²	
P_v	=	299	kN	
$0.6P_v$	=	179	kN	> V = 38.9 kN Low shear, OK !



Site	Dubai Depot TPS (Jebel Ali - Bldg 9) Side Beam Design	Date	July '07	Job No.	068010
		Designed by Rudy	Sheet No.		

Member Properties

Beam depth, D	=	250 mm
Beam width, B	=	250 mm
Flange thickness, T	=	15 mm
Web thickness, t	=	8 mm
Depth btw fillets, d	=	220 mm
Sectional area, Ag	=	9260 mm ²
Moment of inertia, Iz	=	1.11E+08 mm ⁴
Moment of inertia, Iy	=	3.91E+07 mm ⁴
Section Modulus, Zz	=	8.86E+05 mm ³
Section Modulus, Zy	=	3.13E+05 mm ³

* Plastic modulus, Sz	=	9.91E+05 mm ³
* Plastic modulus, Sy	=	3.32E+04 mm ³
[*calculated based on contribution from flanges & web]		

E	=	205000 N/mm ²
v	=	0.3
G	=	$E / 2 * (1 + v)$
	=	78846 N/mm ²
Steel grade	=	50
Strength, py	=	355 N/mm ²

Radius of gyration, rz	=	109.38 mm
Radius of gyration, ry	=	64.96 mm
A,eff	=	9260 mm ²
Zz,eff	=	8.30E+05 mm ³
Zy,eff	=	3.13E+05 mm ³

Section Classification

Epsilon = $[275/py]^{0.5}$	=	0.880
Outstand element of compression flange, b/T		
Plastic	=	7.9
Compact	=	8.8
Semi-compact	=	13.2
Actual b/T	=	8.066667

Flange is compact.**Section classified as****Compact**

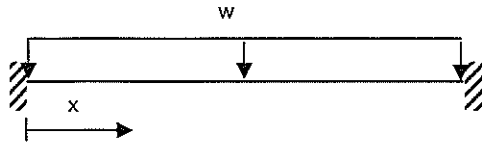
Web subject to compression throughout, d/t		
Axial Force P_{max}	=	0.00 kN
r_1	=	0.00
Plastic	=	70.4
Compact	=	88.0
r_2	=	0.00
Semi-compact	=	105.62
Actual d/t	=	27.50
Web is plastic.		

For symmetrical I & H section

Torsional constant, J	=	$(1/3) * [2BT^3 + (D-2T) * t^3]$	=	6.00E+05
Warping constant, H	=	$0.25 * I_y * (D-T)^2$	=	5.394E+11
Warping static moment, Sws	=	$(D-T) * B^2 * T / 16$	=	1.38E+07
Normalised warping function, Wns	=	$(D-T) * B / 4$	=	1.47E+04



Site	Dubai Depot TPS (Jebel Ali - Bldg 9) Side Beam Design	Date	July '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Loading

$$\begin{aligned} \text{Span Length, } L_x &= 3.2 \text{ m} \\ M_x &= 0.5wL_x^2 \left[\left(\frac{x}{L_x} \right) - \left(\frac{x}{L_x} \right)^2 - \frac{1}{6} \right] \\ e, \text{ eccentricity} &= 350 \text{ mm} \end{aligned}$$

$$F_{y1} - \text{Self wt} = 0.73 \text{ kN/m}$$

F_{y2} - Purlin + Cladding + Insulation

$$\begin{aligned} \text{Height} &= 4.5 \text{ m} \\ \text{Weight} &= 0.5 \text{ kN/m}^2 \\ \text{Load} &= 2.3 \text{ kN/m} \end{aligned}$$

H - Wind Load

$$\begin{aligned} \text{Pressure} &= 1.24 \text{ kN/m}^2 \\ \text{Load} &= 5.6 \text{ kN/m} \end{aligned}$$

For bending along minor axis ($w = 1.4DL$)

$$\begin{aligned} \text{At } x = 0 \text{ and } x = L & \quad M_y(\text{max}) = \frac{wLx^2}{12} = 3.6 \text{ kNm} \\ \text{At } x = 0.25L \text{ and } x = 0.75L & \quad M_y = \frac{wL^2}{96} = 0.4 \text{ kNm} \\ \text{At } x = 0.5L & \quad M_y = \frac{wL^2}{24} = 1.8 \text{ kNm} \end{aligned}$$

$$\begin{aligned} m_{LT} &= 0.2 + (0.15M_2 + 0.5M_3 + 0.15M_4) / M_{\text{max}} = 0.49 \\ m_{LT} * M_{y\text{max}} &= 1.7 \text{ kNm} \end{aligned}$$

For Bending along major axis ($w = H$)

$$\begin{aligned} \text{At } x = 0 \text{ and } x = L & \quad M_z(\text{max}) = \frac{wLx^2}{12} = 4.8 \text{ kNm} \\ \text{At } x = 0.25L \text{ and } x = 0.75L & \quad M_z = \frac{wL^2}{96} = \\ \text{At } x = 0.5L & \quad M_z = \frac{wL^2}{24} = \end{aligned}$$

$$\begin{aligned} m_{LT} &= 0.2 + (0.15M_2 + 0.5M_3 + 0.15M_4) / M_{\text{max}} = 0.20 \\ m_{LT} * M_{z\text{max}} &= 1.0 \text{ kNm} \end{aligned}$$

$$\text{Torsional moment (} w = 1.4F_{y2} \text{)} \quad T_q(\text{max}) = w * e * L_x = 3.5 \text{ kNm}$$

$$\text{Shear in z-direction (} w = 1.4DL \text{)} \quad F_{vx} = \frac{wLx}{2} = 6.7 \text{ kN}$$

$$\text{Shear in y-direction (} w = H \text{)} \quad F_{vy} = \frac{wLx}{2} = 8.9 \text{ kN}$$

$$\text{Resultant shear} \quad F_r = \sqrt{F_{vx}^2 + F_{vy}^2}^{0.5} = 11.1 \text{ kN}$$

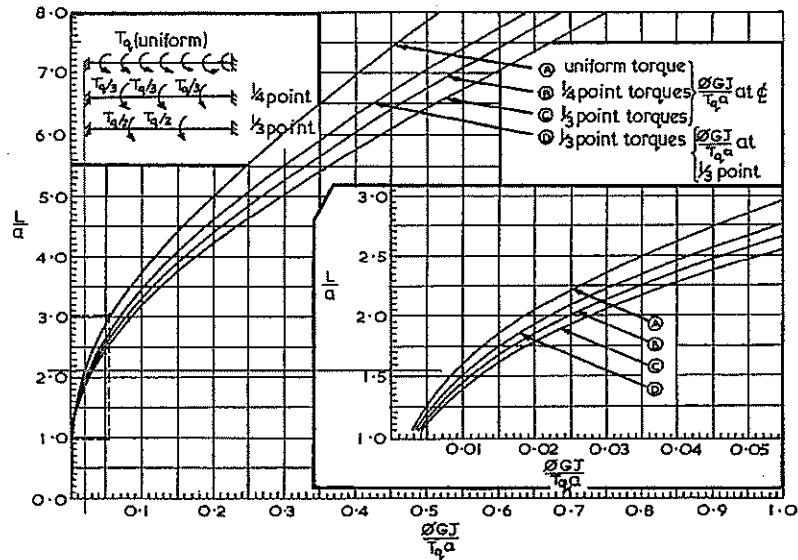
Site Dubai Depot
TPS (Jebel Ali - Bldg 9)
Side Beam Design

Date July '07
Designed by Rudy

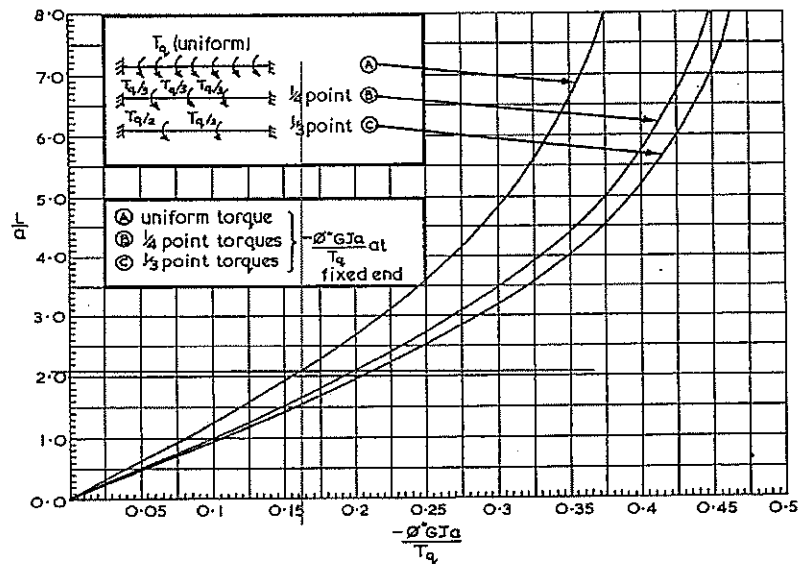
Job No. 068010
Sheet No.

106

GRAPH 7



GRAPH 8



107



Site	Dubai Depot TPS (Jebel Ali - Bldg 9) Side Beam Design	Date	July '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Shear Capacity Check

Shear capacity of beam, $P_v = 0.6p_y A_v = 426 \text{ kN} > \text{Design Value, OK}$

Since d/t is not more than 63 ϵ

Shear buckling resistance check is not required

Clause 4.2.3

Local capacity check

Torsional bending constant, $a = (EH/GJ)^{0.5} = 1528.8477 \text{ mm}$

With uniform torque and $L/a = 2.1$

From the Graph 7

$\phi = 0.02 T_q a / GJ$
 $= 0.002 \text{ rad}$

$Mzt = \phi Mz$
 $= 0.0 \text{ kNm}$

$\sigma_{byt} = M_{yt} / Z_y$
 $= 0.03 \text{ N/mm}^2$

$\sigma_{by} = M_y / Z_y$
 $= 11.38 \text{ N/mm}^2$

From the Graph 8

$\phi'' = 0.165 T_q / GJ a$
 $= 8.05E-09$

$\sigma_w = E W_{ns} \phi''$
 $= 24.2 \text{ N/mm}^2$

$\sigma_{bz} = M_z / Z_z$
 $= 5.37 \text{ N/mm}^2$

$\sigma_{bz} + \sigma_{by} + \sigma_{byt} + \sigma_w = 41.0 \text{ N/mm}^2$ **Ok, less than p_y**

Overall buckling check

$(A_{eff}/A_g)^{0.5} = 1.00$

Effective length, $L_{zz} = 4.4 \text{ m}$ Effective length, $L_{yy} = 4.4 \text{ m}$

Slenderness Ratio: $= 40.6$ Slenderness Ratio: $= 68.35$

Design Lamda, $\lambda = 68.4$

$\beta_w = 1.00$

Torsional index, $x = 16.7$

$\lambda/x = 4.1$

For equal flanges

Buckling parameter, $u = 0.9$

Slenderness factor, $v = 1 / [(1 + 0.05 (\lambda/x)^2)^{0.25}] = 0.86$

Equivalent Slenderness Ratio $\lambda_{LT} = u v \lambda (\beta_w)^{0.5} = 52.8$

Limiting Slenderness Ratio $\lambda_{L0} = 0.4 (\pi^2 E / p_y)^{0.5} = 17$

$\alpha_{LT} = 7$ (Appendix B - Cl 2.2)

$\eta_{LT} = \alpha_{LT} (\lambda_{LT} - \lambda_{L0}) / 1000 = 0.250$

Euler strength, $p_e = (\pi^2 E / \lambda_{LT}^2) = 725.4 \text{ N/mm}^2$

$\phi_{LT} = 0.5 [p_y + (\eta_{LT} + 1) p_e] = 631.0 \text{ N/mm}^2$

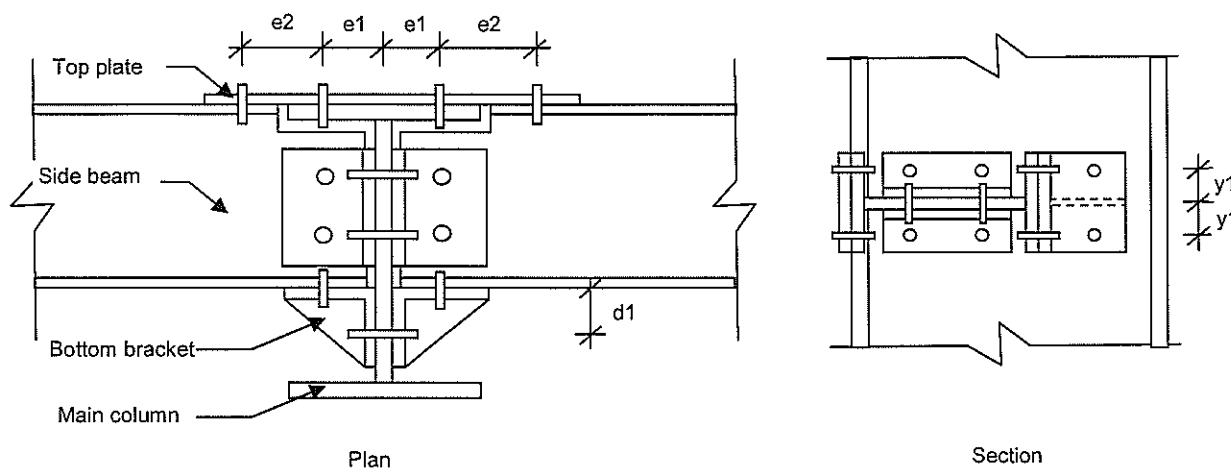
Bending strength, $p_{bz} = p_e p_y / [\phi_{LT} + (\phi_{LT}^2 - p_e p_y)^{0.5}] = 256.0 \text{ N/mm}^2$

Buckling Resis. Momt. $M_{bz} = 253.6 \text{ kNm}$

$m_{LT} M_z / M_{bz} + m_{LT} M_y / p_y Z_y + (\sigma_{byt} + \sigma_w) / p_y [1 + 0.5 (m_{LT} M_y / M_{bz})]$
 $= 0.07$ **Ok, less than 1**

Site	Dubai Depot	Date	July '07	Job No.	068010
	TPS (Jebel Ali - Bldg 9)				
	Side Beam Design	Designed by Rudy		Sheet No.	

Side Beam End Connection Design



From the side beam design

Vertical bending moment	$M_y =$	3.6 kNm
Horizontal bending moment	$M_z =$	4.8 kNm
Torsional bending moment	$T_q =$	3.5 kNm
Vertical shear	$F_y =$	8.9 kN
Horizontal shear	$F_z =$	6.7 kN

Assumption

- 1) Vertical bending moment to be resisted by top and bottom bolt connected to main column
- 2) Horizontal and torsional bending moment to be resisted by bolt connected to side beam's flange
- 3) Vertical shear to be resisted by bolt on the side beam's flange and main column's web
- 4) Horizontal shear to be resisted by bolt on side beam's and main column's web

Side beam dimension

Depth	D1	=	250 mm
Width	B1	=	250 mm
Web thickness	t1	=	15 mm
Flange thickness	T1	=	8 mm

Main column dimension

Depth	D2	=	250 mm
Width	B2	=	250 mm
Web thickness	t2	=	12 mm
Flange thickness	T2	=	8 mm

e1	=	75 mm
e2	=	100 mm
d1	=	0 mm
y1	=	75 mm



Site	Dubai Depot TPS (Jebel Ali - Bldg 9) Side Beam Design	Date	July '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

1) Bolts at the top and bottom of main column's web

a) Tension due to vertical bending moment

$$T = \frac{M_y}{(2 \times y_1 \times \text{no of row of bolt})} = 11.9 \text{ kN} \quad \text{no of row of bolt} = 2$$

b) Shear due to vertical load

$$V_y = \frac{F_y}{\text{no of bolt}} = 2.2 \text{ kN} \quad \text{no of bolt} = 4$$

c) Shear due to horizontal load

$$V_z = \frac{F_z}{\text{no of bolt}} = 1.7 \text{ kN} \quad \text{no of bolt} = 4$$

d) Resultant shear

$$V_r = \left[V_y^2 + V_z^2 \right]^{0.5} = 2.8 \text{ kN}$$

$$\text{Bolt diameter, } \phi = 16 \text{ mm}$$

$$\text{Shear capacity, } V_b = 58.9 \text{ kN}$$

$$\text{Bolt tensile capacity, } T_b = 70.7 \text{ kN}$$

$$T / T_b + V_r / V_b = 0.21 \quad \text{Ok, not greater than 1.4}$$

2) Bolts at the side beam's flange

a) Shear due to horizontal bending moment

$$V_{hs} = \frac{M_z}{(D_1 \times \text{no of row of bolt})} = 9.5 \text{ kN} \quad \text{no of row of bolt} = 2$$

b) Shear due to torsional bending moment

$$V_t = \frac{T_q}{(D_1 \times \text{no of bolt})} = 7.1 \text{ kN} \quad \text{no of bolt} = 2$$

c) Shear due to vertical load

$$V_y = \frac{F_y}{\text{no of bolt}} = 4.5 \text{ kN} \quad \text{no of bolt} = 2$$

d) Resultant shear

$$V_r = \left[V_{hs}^2 + (V_t + V_y)^2 \right]^{0.5} = 14.9 \text{ kN}$$

$$\text{Bolt diameter, } \phi = 16 \text{ mm}$$

$$\text{Shear capacity, } V_b = 58.9 \text{ kN} \quad \text{Ok}$$

3) Bolt at the side beam's web

a) Shear due to horizontal load

$$V_z = \frac{F_z}{\text{no of bolt}} = 3.3 \text{ kN} \quad \text{no of bolt} = 2$$

$$\text{Bolt diameter, } \phi = 16 \text{ mm}$$

$$\text{Shear capacity, } V_b = 58.9 \text{ kN} \quad \text{Ok}$$

4) Bolts at the bracket connected to main column's web

a) Tension due to horizontal bending moment

$$T = \frac{M_z}{[(D_1 + d_1) \times \text{no of row of bolt}]} = 9.5 \text{ kN} \quad \text{no of row of bolt} = 2$$

$$\text{Bolt diameter, } \phi = 16 \text{ mm}$$

$$\text{Bolt tensile capacity, } T_b = 70.7 \text{ kN} \quad \text{Ok}$$

4 CONNECTION DESIGN

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Rev A4

Date: 21 Jul 2007

Page

Calculation/Sketch

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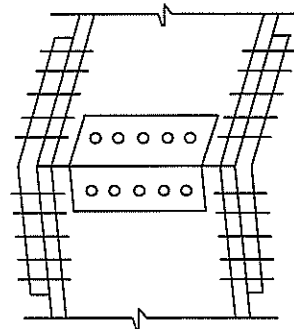
Site	Dubai Depot - Jebel Ali TPS (Building 9)	Date	Jul '07	Job No.	068010
Designed by Rudy			Sheet No.		

Column/Beam Splice Connection Design

From the STAAD Pro analysis

Moment, M_z	=	15.6 kNm
Shear, F_y	=	14.5 kN
Axial Force, F_x	=	37.8 kN

Beam -	depth (D)	=	250 mm
	width (B)	=	250 mm
	web thickness (tw)	=	8 mm
	flange thickness (tf)	=	12 mm



Assume that the direct horizontal and vertical shear are resisted by bolt on the beam web

Resultant shear, V_s	=	40 kN
No of row of bolt provided	=	1
No of bolt provided	=	2
Shear per bolt	=	20 kN
Bolt diameter	=	16 mm
Bolt's shear capacity	=	58.9 kN

Assume shear due to bending is resisted by bolt on the beam flanges

Shear on bolt on the top and bottom flange due to Moment

Eccentricity of vertical load, e	=	50 mm
Moment due to eccentricity, M_e	=	$F_y \cdot e$ 0.7 kNm

Total moment, M_t	=	$M_e + M_z$
	=	65.6 kNm

Shear on the bolt at top and bottom flange	=	$M_t / (D - t_f)$
	=	276 kN
No of row of bolt provided	=	2
No of bolt provided	=	2
Shear per bolt	=	69 kN
Bolt diameter	=	20 mm
Bolt's shear capacity	=	91.9 kN

Calculation/Sketch

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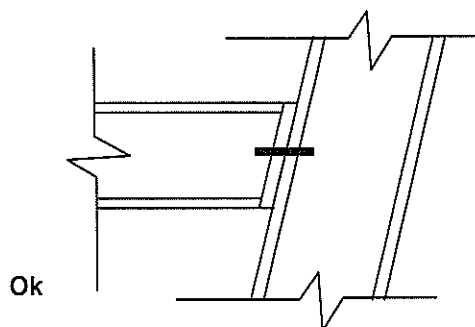
Site	Dubai Depot - Jebel Ali TPS (Building 9)	Date	Jul '07	Job No.	068010
Designed by Rudy				Sheet No.	

Cantilever Beam Connection Design

From the STAAD Pro analysis (Beam 11)

1) At the location connecting to side frame

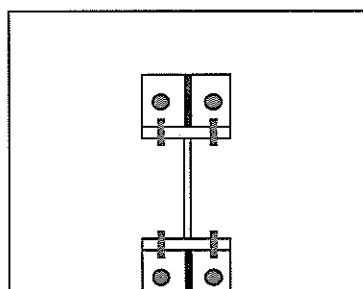
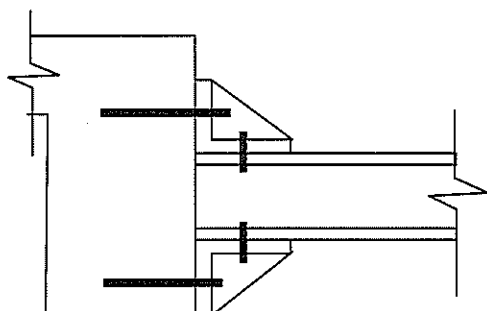
Shear, F_y	=	1.42 kN
No of row of bolt provided	=	1
No of bolt provided	=	2
Shear per bolt	=	1 kN
Bolt diameter	=	16 mm
Bolt's shear capacity	=	58.9 kN



Weld size, s	=	4 mm
Weld strength, pw	=	0.700 kN/mm
Weld length required	=	F_y/pw
	=	2
Weld length provided	=	1000

mm
mm OK

2) At the location connecting to roof beam / slab



From the STAAD Pro analysis (Beam 11)

Shear, F_y	=	2.37 kN
Moment, M_z	=	2.52 kNm
Axial Force, F_x	=	31.7 kN

a) Design of bolt at the beam top and bottom flange

Depth of beam, D	=	250 mm
Flange thickness, T	=	12 mm

Shear on the bolts, V	=	$M_z/(D-T) + F_x$
	=	42.3 kN
No of bolt provided	=	2
Shear per bolt	=	21 kN
Bolt diameter	=	16 mm
Bolt's shear capacity	=	58.9 kN

Ok

Calculation/Sketch

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Site	Dubai Depot - Jebel Ali TPS (Building 9)	Date	Jul '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

b) Design of bolt connecting to RC Roof Slab / Beam

$$\begin{aligned}\text{Distance between bolt, } db &= 300 \text{ mm} \\ \text{Tension force on the bolt, } T &= Mz/Db + Fx \\ &= 40.1 \text{ kN}\end{aligned}$$

$$\begin{aligned}\text{No of bolt provided} &= 2 \\ \text{Tension force per bolt, } T1 &= 20 \text{ kN} \\ \text{Bolt diameter} &= 16 \text{ mm} \\ \text{Bolt's tension capacity, } Tb &= 70.3 \text{ kN}\end{aligned}$$

$$\begin{aligned}\text{Shear force per bolt, } V1 &= Fy / \text{no of bolt} \\ &= 1.2 \text{ kN} \\ \text{Bolt diameter} &= 16 \text{ mm} \\ \text{Bolt's shear capacity, } Vb &= 58.9 \text{ kN}\end{aligned}$$

$$V1 / Vb + T1 / Tb = 0.31 \text{ Ok, not greater than 1}$$

c) Check weld connecting the stiffener plate to the angle

$$\begin{aligned}\text{Weld size, } s &= 4 \text{ mm} \\ \text{Weld strength, } pw &= 0.700 \text{ kN/mm} \\ \text{Weld length required} &= \max[(T+Fy), V] / pw \\ &= 60.7 \text{ mm} \\ \text{Weld length provided} &= 284 \text{ mm} \quad \text{OK}\end{aligned}$$

Calculation/Sketc

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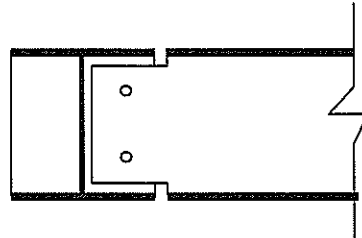
Site	Dubai Depot - Jebel Ali	Date	Jul '07	Job No.	068010
	TPS (Building 9)				
	Side Frame to Roof Beam Connection	Designed by	Rudy	Sheet No.	

1) At the end of the frame

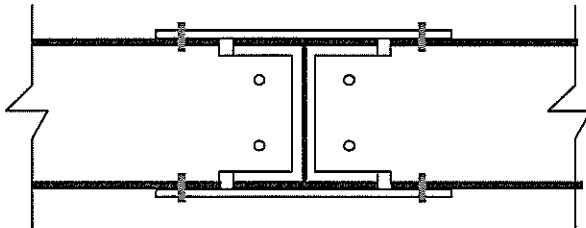
From the STAAD Pro analysis (beam 6)

Shear, Fy	=	14.8 kN
Moment, Mz	=	0 kNm
Axial Force, Fx	=	1.74 kN
No of bolt provided	=	2
Shear per bolt	=	7 kN
Bolt diameter	=	16 mm
Bolt's shear capacity	=	58.9 kN

Ok



2) At the end column support



From the STAAD Pro analysis (beam 6)

Shear, Fy	=	19.8 kN
Moment, Mz	=	9.0 kNm
Axial Load, Fx	=	4.7 kN

Assume the moment is resisted by bolts at the top and bottom flange

Beam depth, D	=	250 mm
Flange thickness, T	=	8 mm
Shear on the bolt, Vs	=	$M/(D-T)$
	=	37.2
No of row of bolt	=	1
No of bolt provided	=	2
Shear per bolt	=	19 kN
Bolt diameter	=	16 mm
Bolt's shear capacity	=	58.9 kN

Ok

Assume direct shear and axial load is resisted by bolts at the beam web

No of bolt provided	=	2
Shear per bolt	=	10 kN
Bolt diameter	=	16 mm
Bolt's shear capacity	=	58.9 kN

Ok

Calculation/Sketc

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Site	Dubai Depot - Jebel Ali TPS (Building 9)	Date	Jul '07	Job No.	068010
	Beam Splice Connection	Designed by	Rudy	Sheet No.	

DL

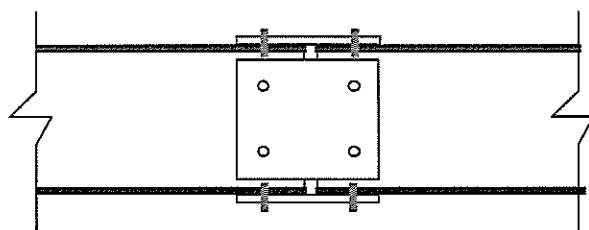
Selfweight	0.40 x	1.00 =	0.4
Sheeting	0.10 x	3.80 =	0.38
Purlins	0.10 x	3.80 =	0.38
Insulation	0.30 x	3.80 =	1.14
M&E Finishes	0.30 x	3.80 =	1.14
			<u>3.44 kN/m</u>

<u>LL</u>	0.60 x	3.80 =	2.28 kN/m
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$w(uls) = 8.5 \text{ kN/m}$

span L = 3.6 m
Mmax = 13.7 kNm
Vmax = 15.2 kN

Beam splice



Assume the moment is resisted by bolts at the top and bottom flange

Beam depth, D	=	250 mm	
Flange thickness, T	=	8 mm	
Shear on the bolt, Vs	=	$M/(D-T)$	
	=	56.7	
No of row of bolt	=	1	
No of bolt provided	=	2	
Shear per bolt	=	28 kN	
Bolt diameter	=	16 mm	
Bolt's shear capacity	=	58.9 kN	Ok

Assume direct shear is resisted by bolts at the beam web

No of bolt provided	=	2	
Shear per bolt	=	8 kN	
Bolt diameter	=	16 mm	
Bolt's shear capacity	=	58.9 kN	Ok

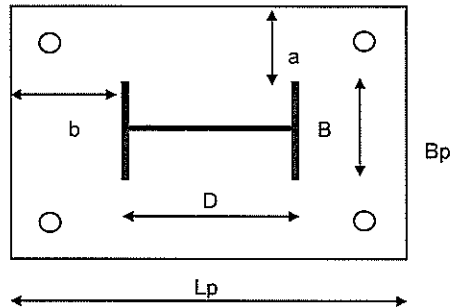
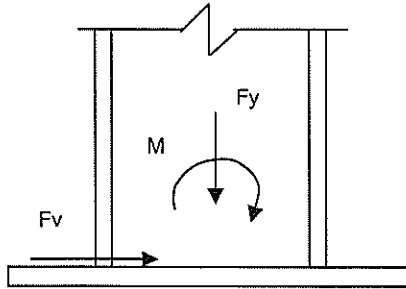
Calculation/Sketch

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Site	Dubai Depot	Date	Jul '07	Job No.	068010
	TPS (Jebel Ali B9)				
	Ground Level	Designed by	Rudy	Sheet No.	

Base Plate Design



Applied Load

Fy	=	37.7 kN
Fv	=	15.21 kN

Column section

Depth, D	=	250 mm
Width, B	=	250 mm
Flange thickness, tf	=	12 mm
Web thickness, tw	=	8 mm

Base plate

Length, Lp	=	350 mm
Width, Bp	=	350 mm
Thickness, tp	=	20 mm
pyp	=	345 N/mm ²
fcu	=	40 N/mm ²
0.4 fcu	=	16 N/mm ²

Base plate area required =	2356.25 mm ²
Base plate area provided =	122500 mm ²
Base pressure, w =	0.30776 N/mm ²

a	=	50 mm	b	=	50 mm
Min. plate thickness required	=	$[2.5 \cdot w \cdot (a^2 - 0.3b^2) / pyp]^{0.5}$			
	=	2.0 mm			Ok

Holding down bolt design

No of bolt provided	=	4
Maximum tension per bolt due to axial load, T1	=	0.0 kN
Shear per bolt, V1	=	3.8 kN

Column to base plate weld design

Weld size, s	=	6 mm
Weld strength, pw	=	1.050 kN/mm
Weld length required	=	$(Fy + Fv) / pw$
	=	50 mm
Weld length provided	=	1000 mm
		OK

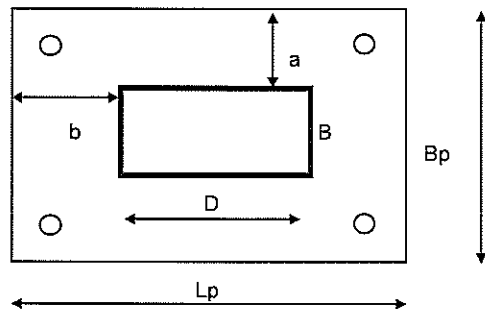
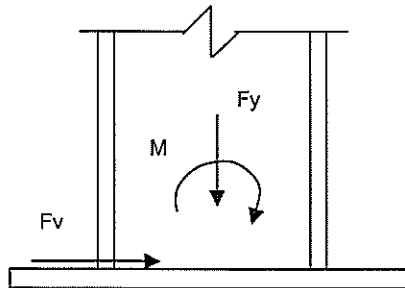
Calculation/Sketch

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Site	Dubai Depot	Date	July '07	Job No.	068010
	TPS (Jebel Ali B9)				
	Roof Level	Designed by	Rudy	Sheet No.	

Base Plate Design



Applied Load

F_y	=	42.18 kN
F_v	=	32.94 kN
M	=	2.48 kNm

Column section

Depth, D	=	150 mm
Width, B	=	150 mm
Flange thickness, t_f	=	6 mm
Web thickness, t_w	=	6 mm

Base plate

Length, L_p	=	350 mm
Width, B_p	=	350 mm
Thickness, t_p	=	20 mm
p_{yp}	=	345 N/mm ²
f_{cu}	=	30 N/mm ²
$0.4 f_{cu}$	=	12 N/mm ²

Holding down bolt design

Distance between the outer bolts, db	=	350 mm
No of bolt provided at the outer line	=	2
Total no of bolt	=	4
Maximum tension per bolt due to bending and axial load, T_1	=	45.7 kN
Shear per bolt, V_1	=	8.2 kN

Using grade 8.8 bolt,

Bolt size	=	16 mm
Tension capacity, T_b	=	70.7 kN
Shear capacity, V_b	=	58.9 kN

$$T_1 / T_b + V_1 / V_b = 0.79 \leq 1.4, \text{ Ok}$$

Column to base plate weld design

Assuming moment is resisted by weld at the column flange

Force on the weld, T_2	=	$M / (D_c - t_f)$
	=	17.2 kN

Effective length of the weld resisting T_2	=	$2 \times B$
	=	300 mm

Weld size, s =	4 mm	(up to 15mm thick)
Weld strength, p_w =	0.7 kN/mm	(based on longitudinal capacity)

Total weld resistance	=	210.0 kN	Ok
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Minimum thickness of plate required, t_p min = $[6M1 / 1.2 pyp]^{0.5}$
 = 8.1 mm Ok